

Functional Genomics, Gene Chips and Proteomics

Further Readings:

Nature 405: 819-846

“Functional Genomics” by S.P. Hunt and F.J. Liversey, Ch. 2-7, 10

“Differential Display Methods and Protocols” by P. Liang and A.B. Pardee, Ch. 1-5

“The Yeast Two-Hybrid System” by P.L. Bartel and S. Fields, Ch. 2 and 7

Acknowledgements:

Dr. KK Wong (Baylor College of Medicine)

Dr. HC Leung (Singapore Genome Institute)

What have already been achieved?

- Completed sequences of more than 20 prokaryotic genomes (e.g. *E. coli*, *Methanococcus jannaschii*, *Mycobacterium tuberculosis*, etc.)
- Yeast, worm, fruit fly
- Human
- *Arabidopsis thaliana* and rice

Impacts of Human Genome Project

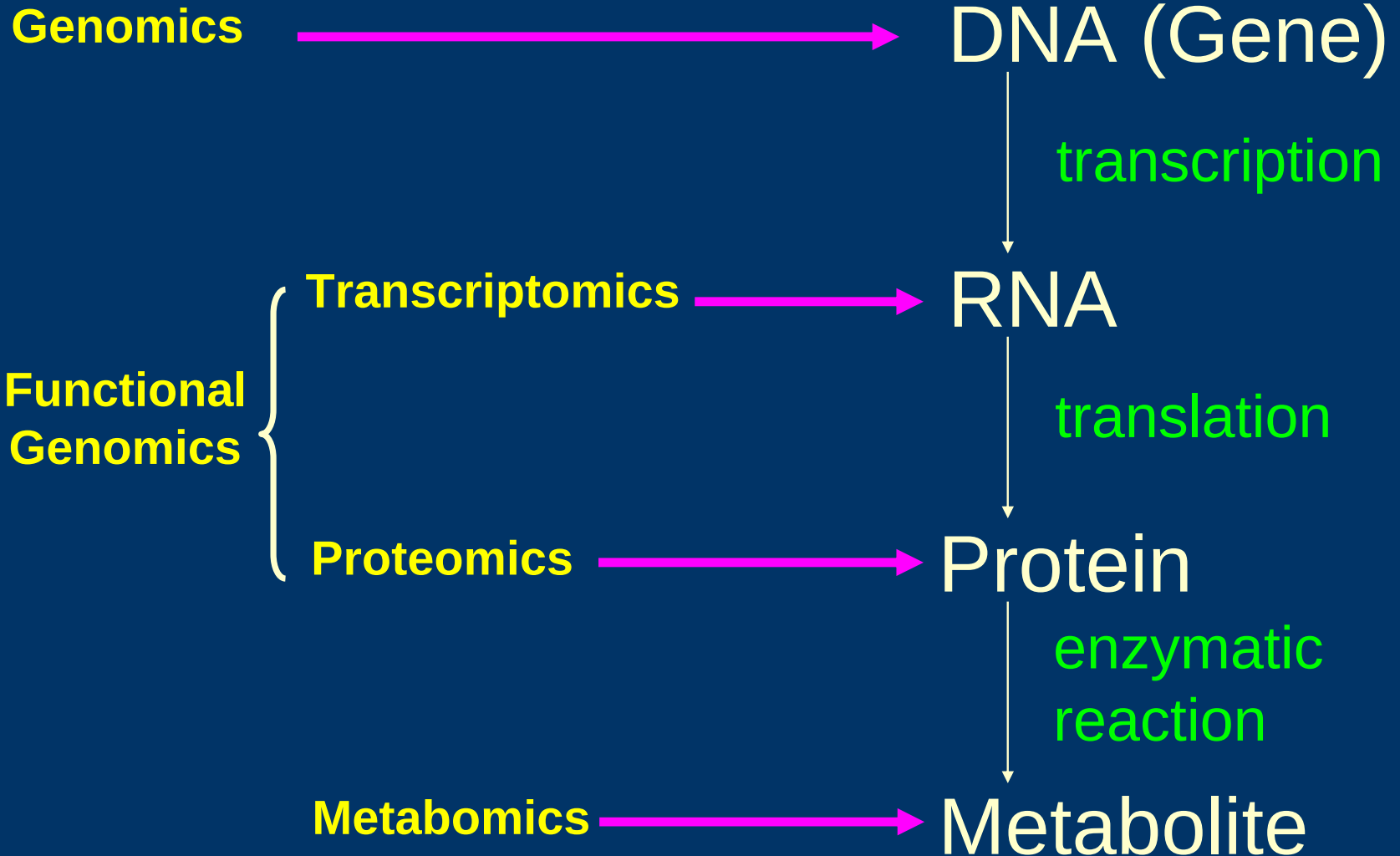
- Led to two major technological breakthroughs:
 - high-throughput DNA sequencing, and
 - powerful computational tools for sequence analysis
- Switch to data-intensive mode of research
- Broad scope and scale
- From one-gene-at-a-time (reductionist) approach to whole-genome approach
- Can keep track of all the components at once i.e. profiling

Current and Potential Applications

- Molecular medicine
- Microbial genomics
- Risk assessment
- Bioarchaeology, anthropology, evolution, and human migration
- DNA forensics (identification)
- Agriculture, livestock breeding, and bioprocessing

What's Next?

- Determine the meaning behind these genetic codes
- Determine the functions of the genes
- **FUNCTIONAL GENOMICS**



Gen

Transcript

Prote

Metab

~ ome =

Sequence of a
complete set of

genes

transcripts

proteins

metabolites

Gen

Prote

~ omics =

Analysis of the

genome

proteome

Structural versus Functional Genomics

- **Structural genomics:**
 - Construction of high resolution genetic, physical and transcript maps of an organism
- **Functional genomics:**
 - Development and application of global experimental approaches to assess gene function by making use of the information and reagents provided by structural genomics

Possible Protein Functions

- Structural component
- Enzyme
- Hormone
- Regulator

Different Approaches to Identify Functions

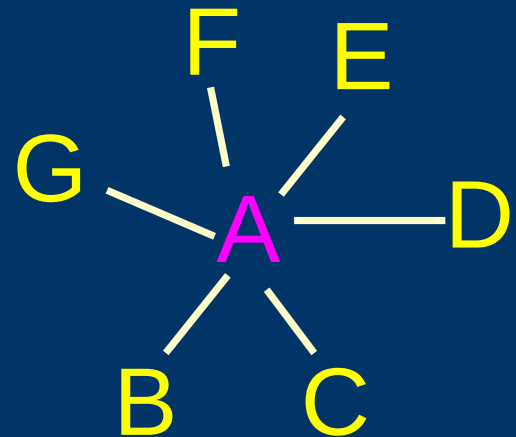
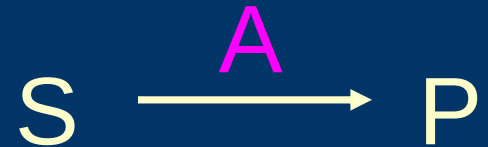
- Purify the gene product (or express the gene product in a heterologous system) and then perform biochemical studies (e.g. enzymatic activities; reaction kinetics; interactions of molecules)
- Perform computational analysis of the gene product (e.g. searching for homologue (orthologues and paralogues) or conserved functional domains; prediction of 3-D structure based on physical parameters)
- Study the physiological effects of the gene product or the compound formed by the action of the gene product (e.g. exogenous application of purified target compounds to cell cultures, tissues, or whole organism)

Different Approaches to Identify Functions

- Gene expression analysis
 - when and where a gene is produced (e.g. differential gene expression; *in situ* hybridization)
 - when and where a protein is produced and located (e.g. differential protein production by 2-D gel electrophoresis, mass spectrometry; immunodetection)
- Gain-of-function and loss-of-function approaches (e.g. construction of overexpressors and underexpressors, RNAi)
- Reverse genetics (e.g. screen of transposon-tagged mutant pools)

Re-Defining “Functions”

- **Molecular function**
 - the function of a protein is defined by its action on substrates to form products
- **Contextual (Cellular) function**
 - the context of a protein interacting with other proteins



From DNA Sequences to Functions

- Expression profiles (transcriptome)
- Protein profiles (proteomics)
- Protein-protein interactions
- Computational analyses and modeling

Expression Profiling (Transcriptome)

- mRNA makes protein
- Determining gene expression pattern will help understanding the biological activities and roles of its encoded protein
- Multi-gene patterns of expression give hint to common regulatory mechanisms
- Genes with similar expression behavior are likely to be related functionally – “guilt-by-association”

To Identify and Isolate Differentially Expressed Genes

- Differential hybridization
- Differential display
- RAP (RNA Arbitrariness Primed)-PCR
- Subtraction library
- SSH (Suppression subtractive hybridization)
- RDA (Representational difference analysis)
- SAGE (Serial Analysis of Gene Expression)
- DNA microarray (membranes) and DNA chip (glass slides)

Differential Hybridization

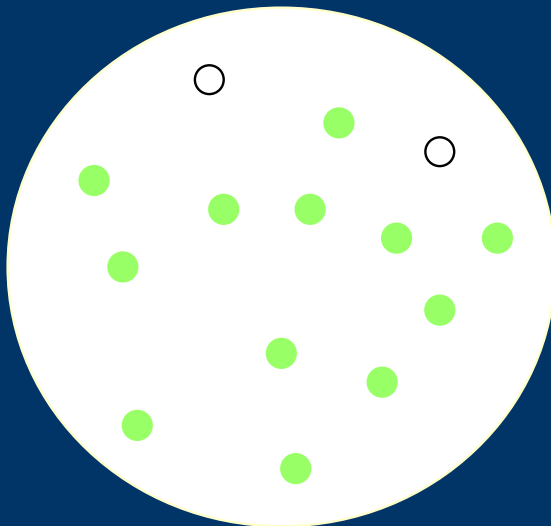
mRNA from sample 1



labeled probes



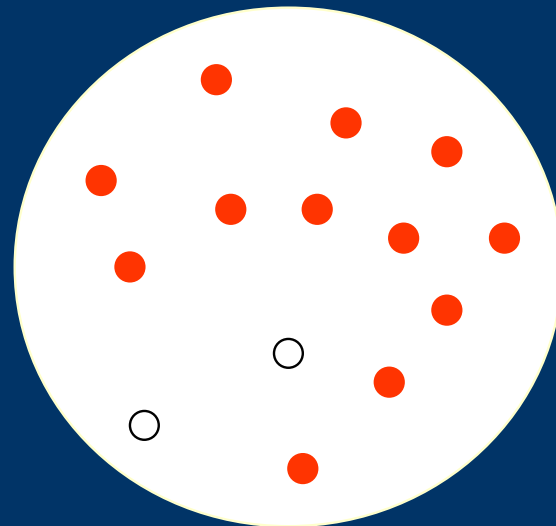
Hybridize with cDNA libraries (made from pooled mRNA of sample 1 and 2) on replicated membranes



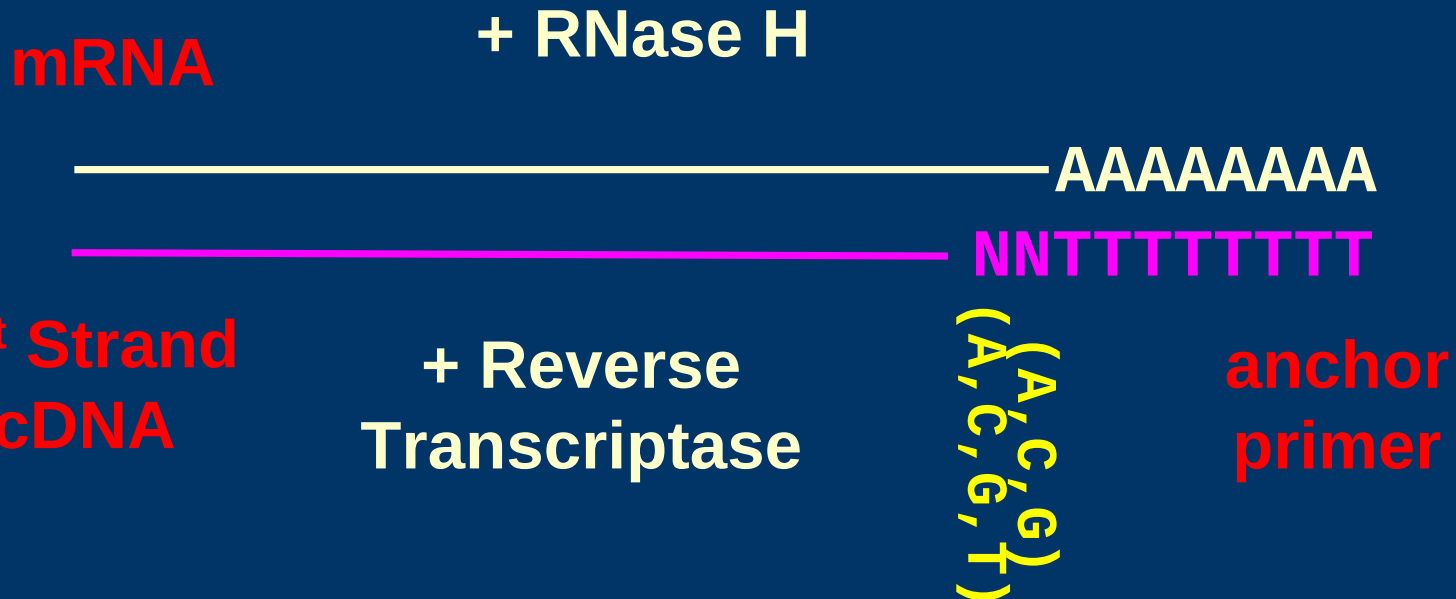
mRNA from sample 2



labeled probes



Differential Display



**Arbitrary
Primer (10-mer)** **Taq, dNTPs** **2nd Strand
cDNA**

NNNNNNNNNN

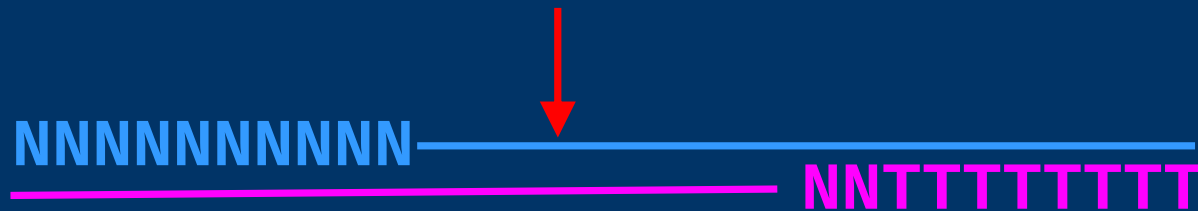
NNNNNNNNNN

**1st Strand
cDNA**

(A', C', G', T')

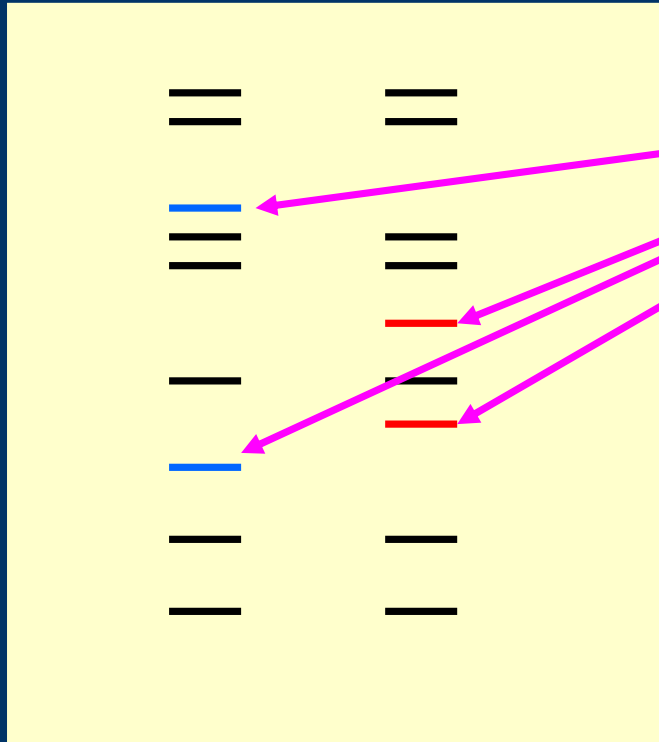


+ Anchor primer
+ Arbitrary primer
+ dNTPs
+Taq
Annealing at 36-38°C



Polyacrylamide Gel

Sample A Sample B



**Cloned for
further analysis**

RAP-PCR

+ RNase H

mRNA



1st Strand
cDNA

Arbitrary
Primer (18-mer)

+ Reverse Transcriptase
(temperature may vary from 36-50°C)

**2nd Strand
cDNA**

NNNNNNNN

NNNNNNNN

**1st Strand
cDNA**

**+ Taq, dNTPs, Arbitrary Primer
Low Annealing Temperature (36-38°C)**

NNNNNNNNN—
—NNNNNNNNN

+ Arbitrary primer

+ dNTPs

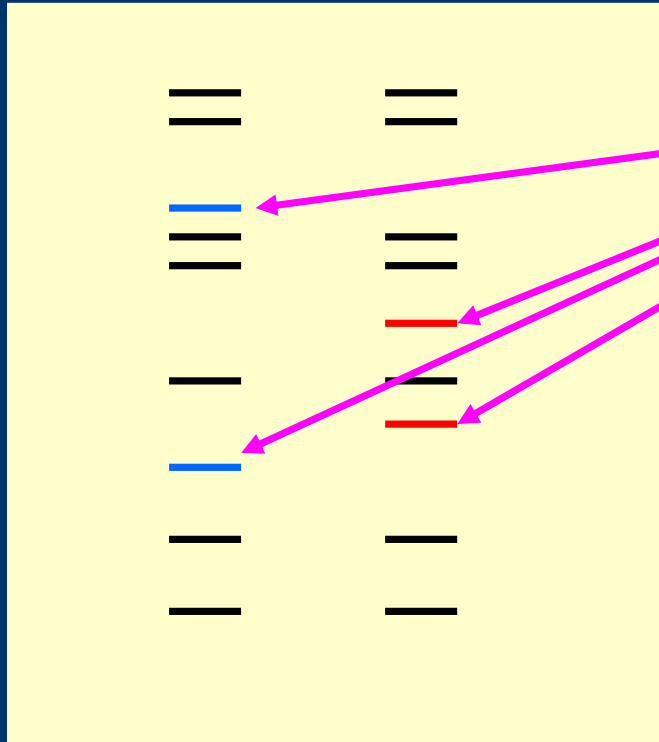
+Taq

(back to regular annealing
temperature after 1-2 initial cycles
using low temperature)

NNNNNNNNN—
—NNNNNNNNN

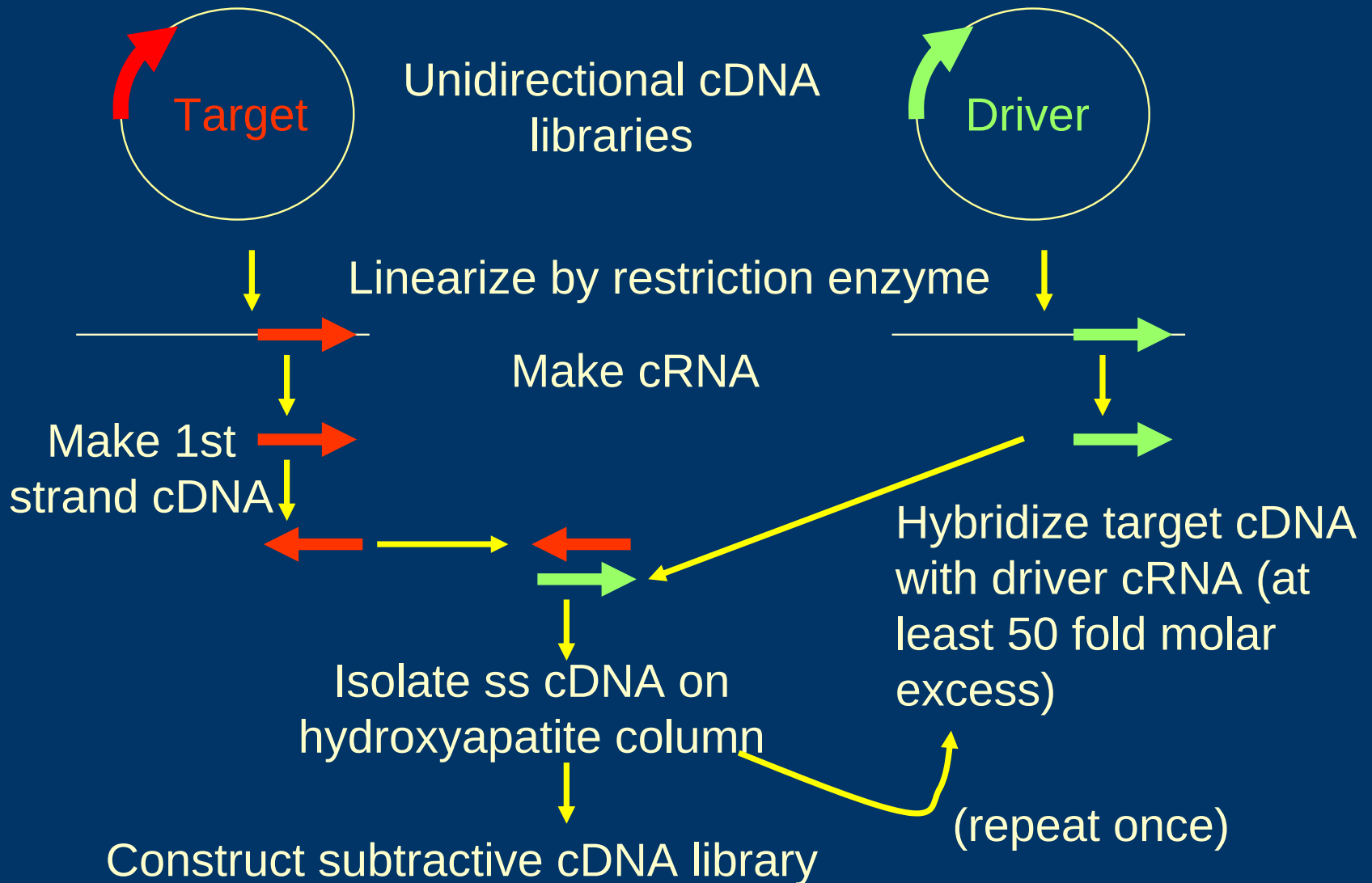
Polyacrylamide Gel

Sample A Sample B

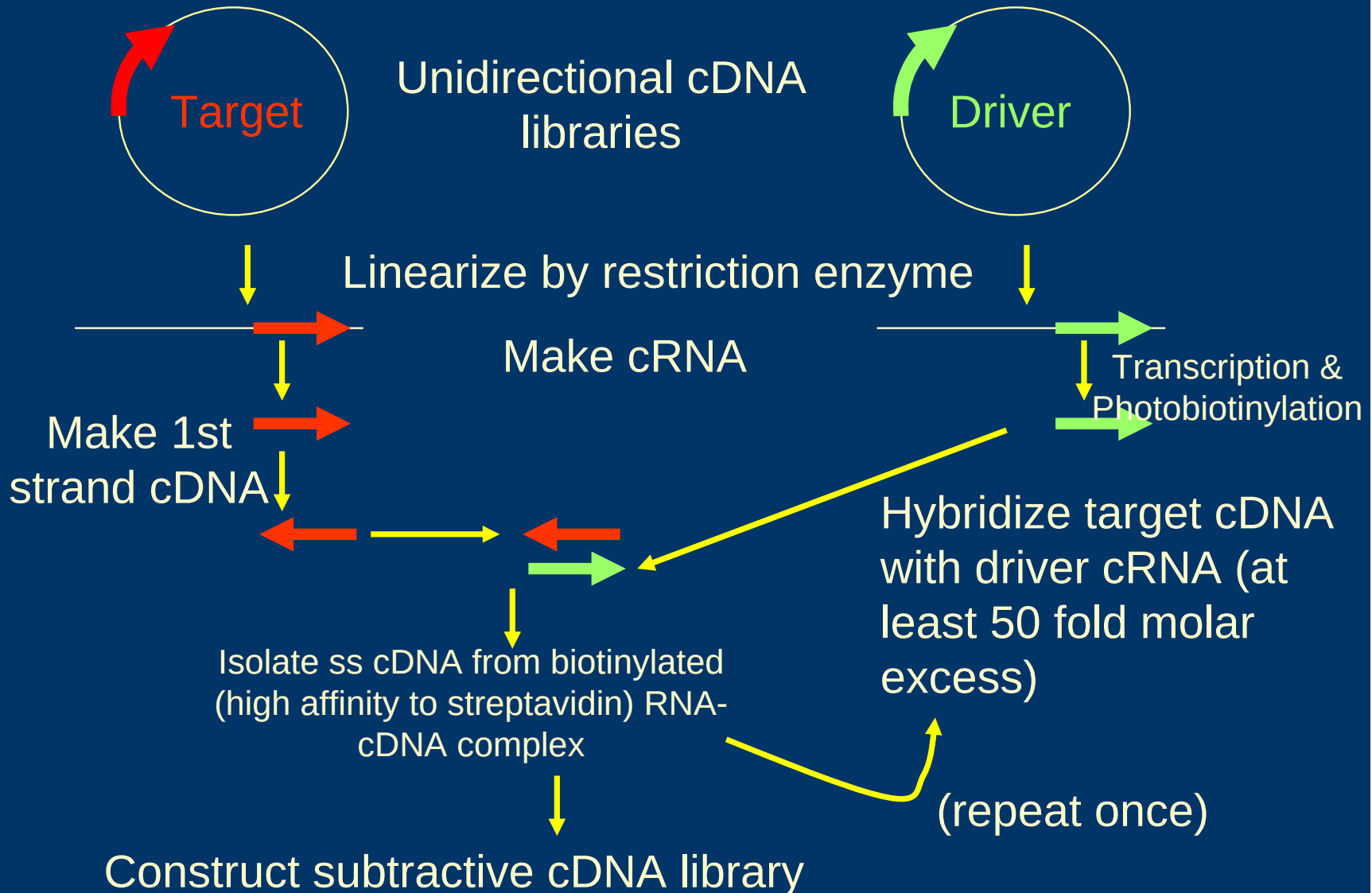


**Cloned for
further analysis**

Subtraction Library (by Hydroxyapatite Column)



Subtraction Library (by Biotinylation)



Suppression Subtractive Hybridization (e.g. Clontech)

Tester mRNA



Tester cDNA



Ligate to
adapter 1



Driver mRNA



Driver cDNA



Cut with *RsaI* to
get shorter,
blunt-ended molecules



No ligation

Tester mRNA



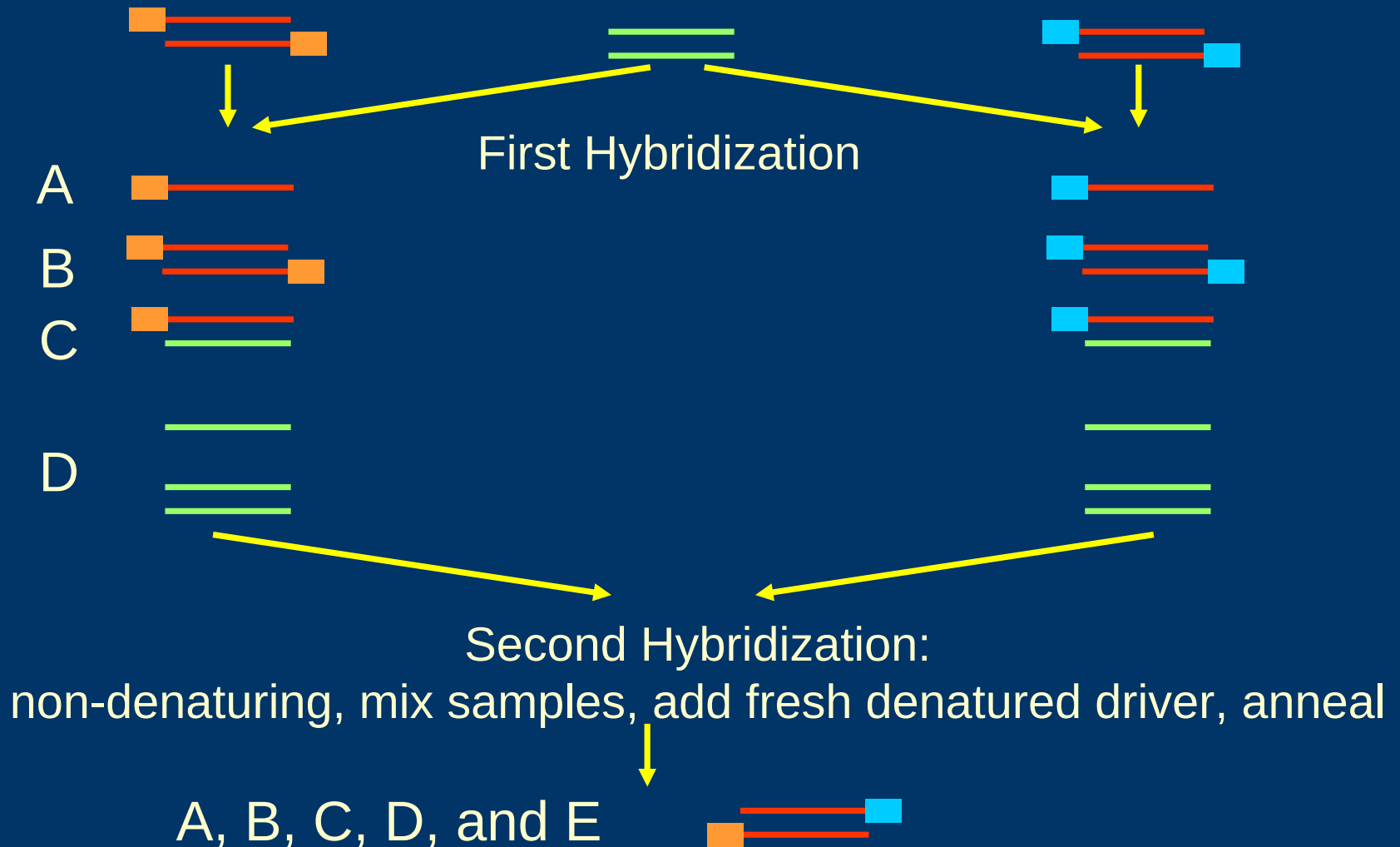
Tester cDNA



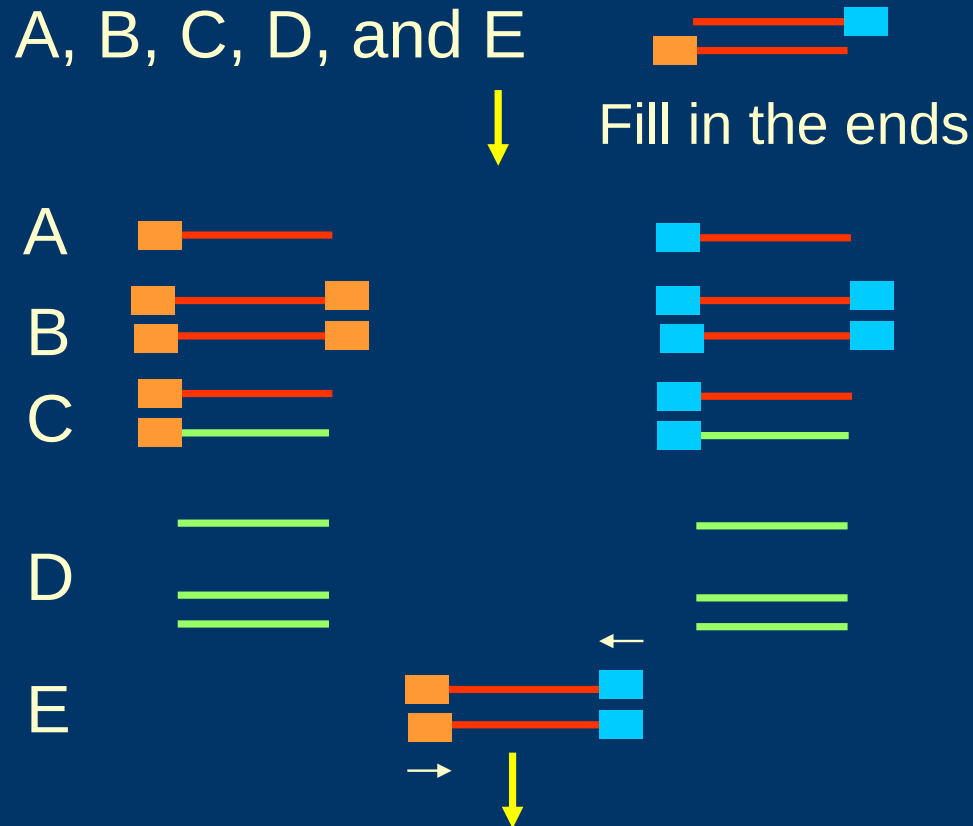
Ligate to
adapter 2



Suppression Subtractive Hybridization (e.g. Clontech)

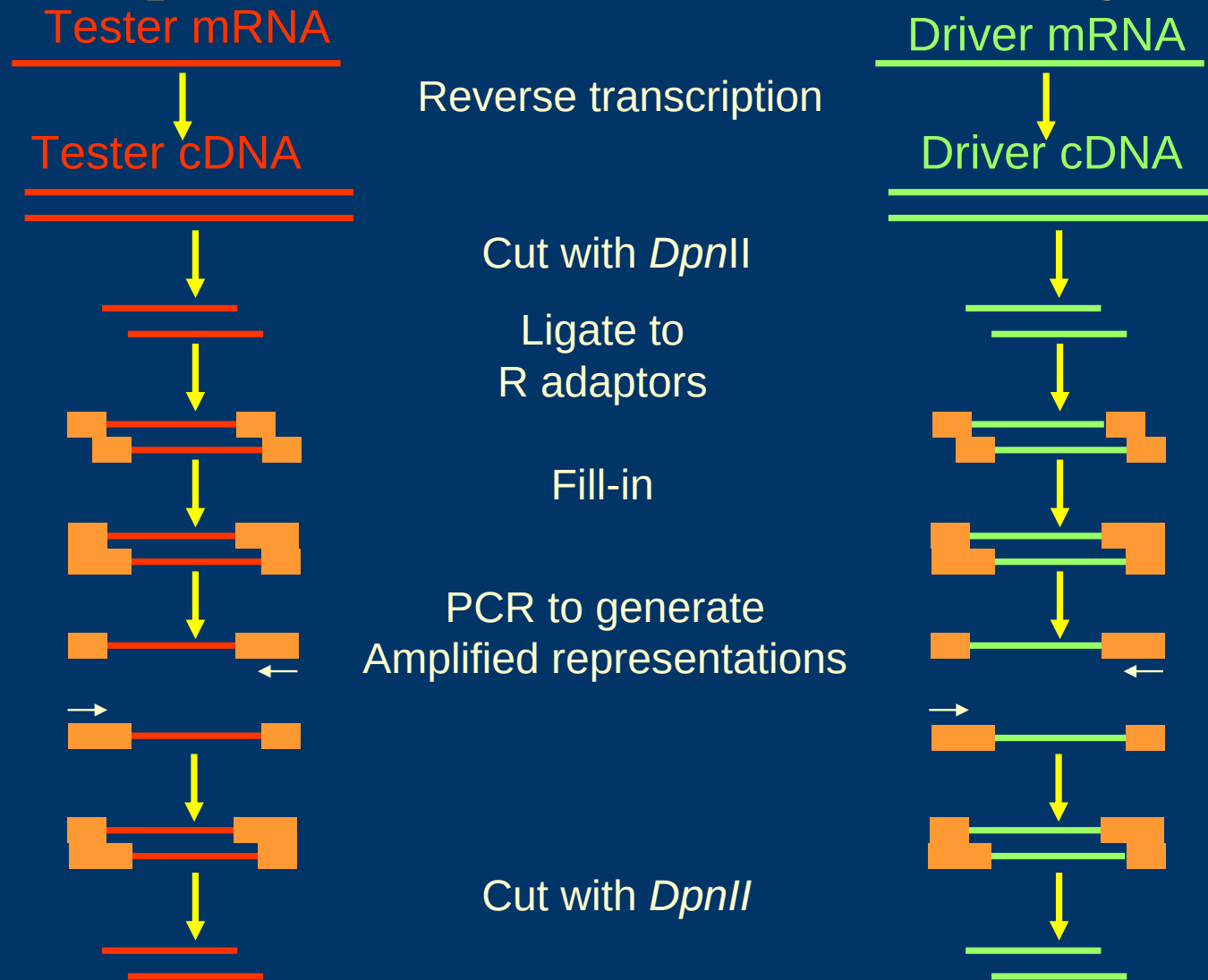


Suppression Subtractive Hybridization (e.g. Clontech)



- Use suppression PCR (anneal at 65°C), only E can give PCR products;
- Use nested primers to enhance specificity

Representational Difference Analysis



Representational Difference Analysis

Tester cDNA

Driver cDNA

Only tester will be ligated to
J adaptors

Denature and hybridization;
driver in excess

Fill-in

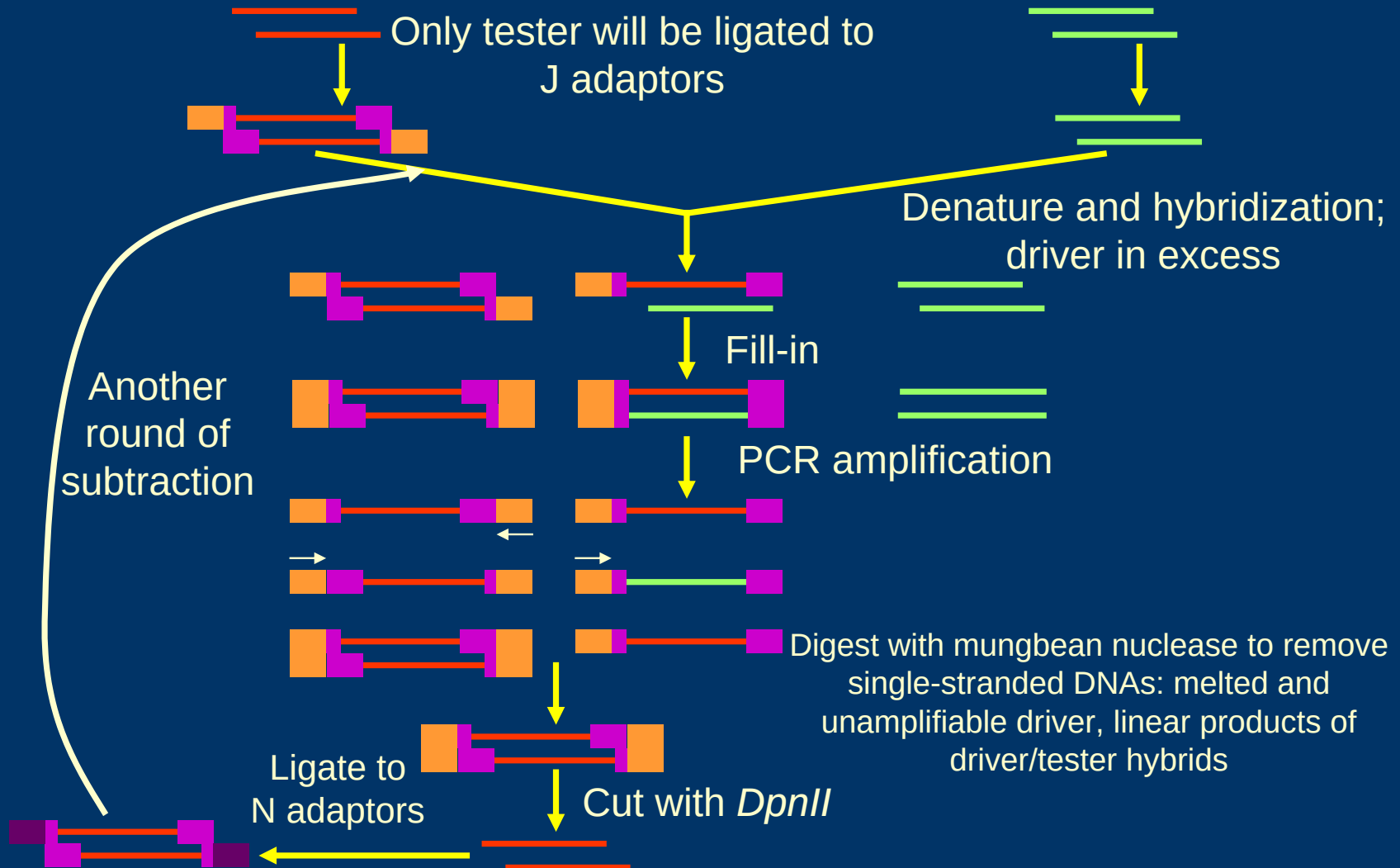
PCR amplification

Digest with mungbean nuclease to remove
single-stranded DNAs: melted and
unamplifiable driver, linear products of
driver/tester hybrids

Ligate to
N adaptors

Cut with *DpnII*

Another
round of
subtraction



SAGE (e.g. I-SAGE from Invitrogen)

———— AAAAAA
 TTTTTT ● Mix RNA sample with oligo
 dT magnetic beads

———— AAAAAA
 TTTTTT ● Reverse transcription

———— AAAAAA
 GTAC ——— TTTTTT ● Cut with a 4-bp cutter, e.g.
*Nla*III 5' CATG ↓ 3'
 3' ↑ GTAC 5'

1 CATG ——— AAAAAA
 GTAC ——— TTTTTT ● Divide sample into 2 portions;
 ligate to adapter 1 and 2 (40
 bps each), respectively.

2 CATG ——— AAAAAA
 GTAC ——— TTTTTT ● Adapter 1 and 2 contain a
 type IIS restriction site, e.g.
*Bsm*F1 5'GGGAC(N)₁₀ ↓ 3'
 3'CCC TG(N)₁₄ ↑ 5'

SAGE (e.g. I-SAGE from Invitrogen)

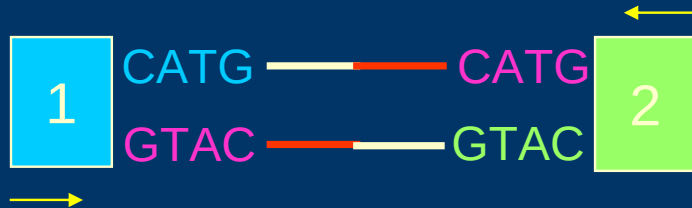


Cut with
*Bsm*F15'GGGAC(N)₁₀↓3'
3'CCC TG(N)₁₄↑5'

Fill in with DNA
polymerase (40 bps
adapter + 14 bps tags)

Ligate to form ditags

SAGE (e.g. I-SAGE from Invitrogen)



PCR amplify



Cut with *Nla*III to release ditags



Ligate to form concatemers

	<u>Sample1</u>	<u>Sample 2</u>
Gene 1	12	14
Gene 2	5	6
Gene 3	30	5
Gene 4	0	23
etc.		

Cloning, sequencing
and statistical analysis
on number of hits

DNA Chips

- Types:
 - oligonucleotides and cDNA
- Key technologies:
 - high density and resolution, precise location arrays; oligo: 250,000 per cm^2 ; cDNA: 10,000 per cm^2
 - laser confocal fluorescence scanning to detect parallel hybridization

DNA Chips

- Advantages:
 - true parallelism
 - multiplexing
 - automation
 - miniaturization

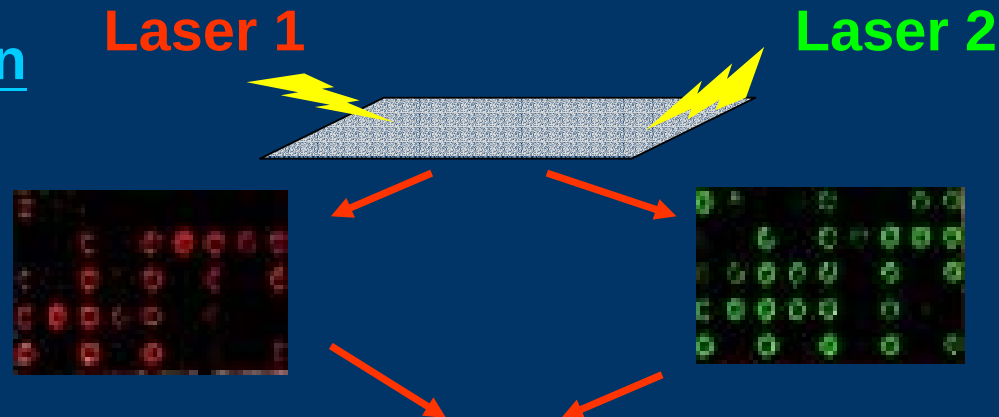


1. Fabrication of arrays

2. Sample preparation

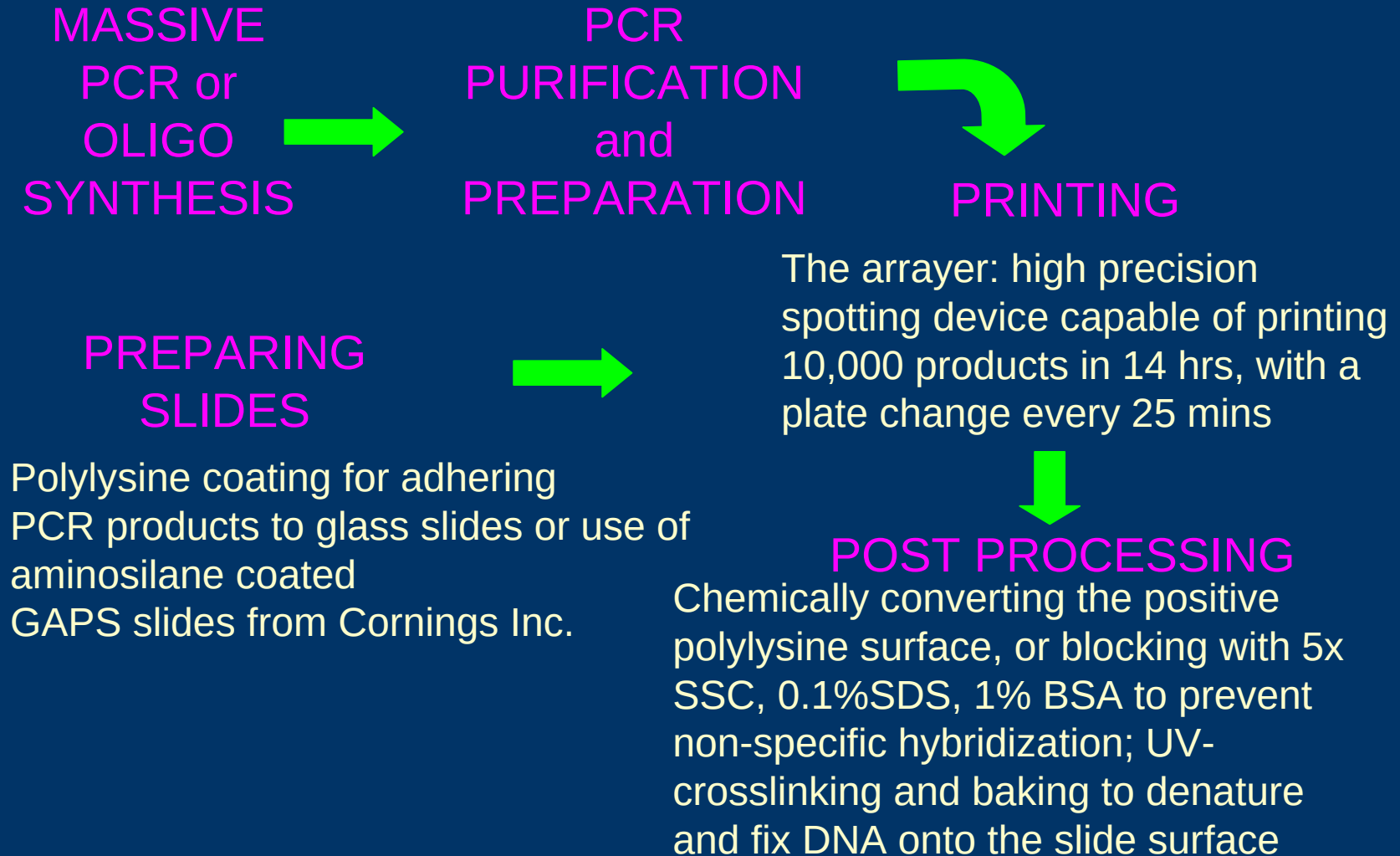
3. Hybridization

4. Detection

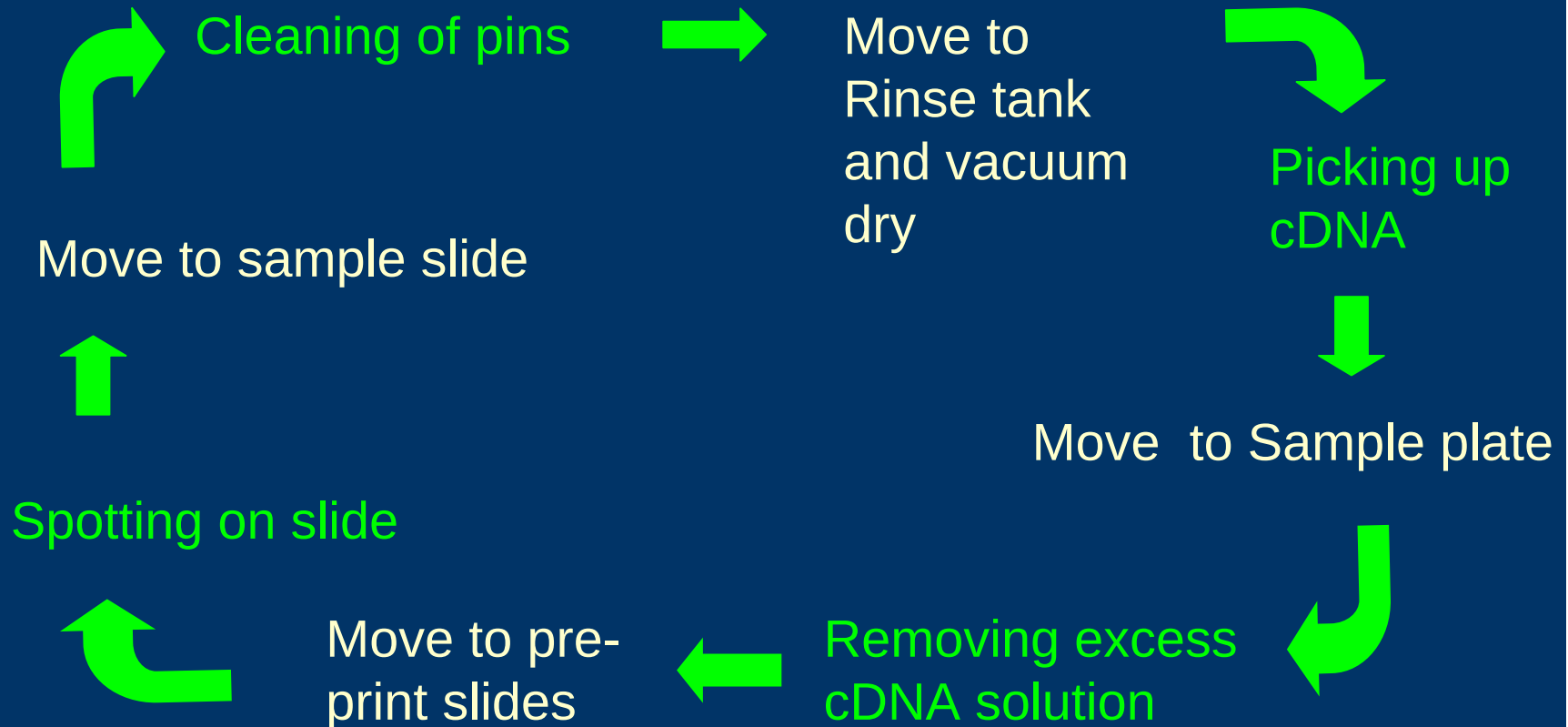


5. Data analysis

Building a Gene Chip

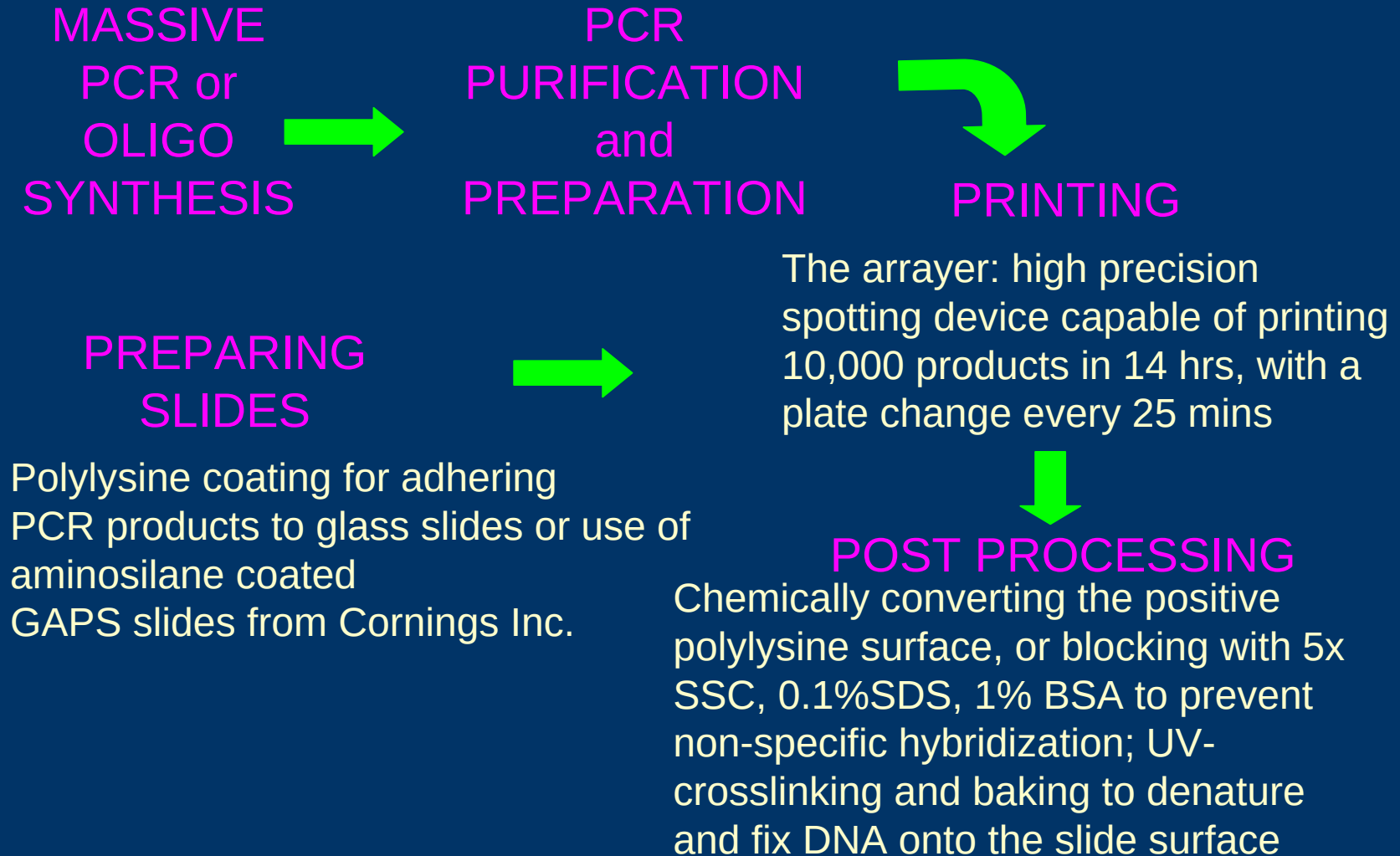


Arraying Process



Making DNA Chips

Building a Gene Chip



Preparation of RNA Probes

COLLECT AND HARVEST SAMPLES

Designing experiments to profile conditions/perturbations/mutations and carefully controlled growth conditions



RNA ISOLATION

RNA yield and purity are determined by system. Two or more RNA samples are hybridized on one chip.



cDNA PRODUCTION

Single strand synthesis or amplification of RNA can be performed. cDNA production includes incorporation of Aminoallyl-dUTP.



Hybridization and Data Analysis

PROBE LABELING

Two RNA samples are labelled with Cy3 or Cy5 monofunctional dyes via a chemical coupling to AA-dUTP. Samples are purified using a PCR cleanup kit.



ARRAY HYBRIDIZATION

Cy3 and Cy5 RNA samples are simultaneously hybridized to chip. Hybs are performed for 5-12 hours and then chips are washed.



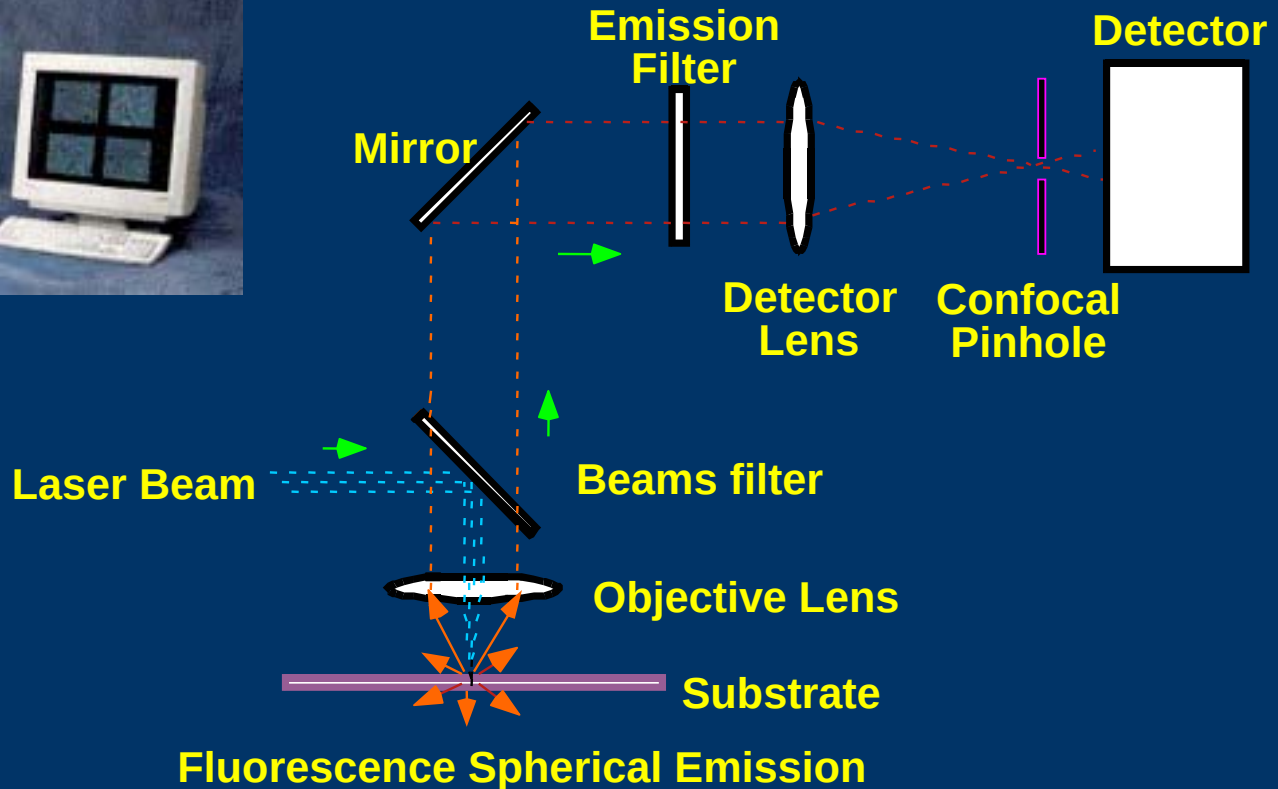
DATA ANALYSIS

Ratio measurements are determined via quantification of 532 nm and 635 nm emission values. Data are uploaded to the appropriate database where statistical and other analyses can then be performed.

A close-up photograph of a metal slide rule. The top part shows a ruler scale with markings from 10 to 80. Below the ruler is a sliding bar with a textured surface. The text "Bake-slides" is overlaid in pink at the bottom.

Bake-slides

Confocal Scanner to Detect Fluorescence Signal from Hybridization Spots



Fluorescent dyes



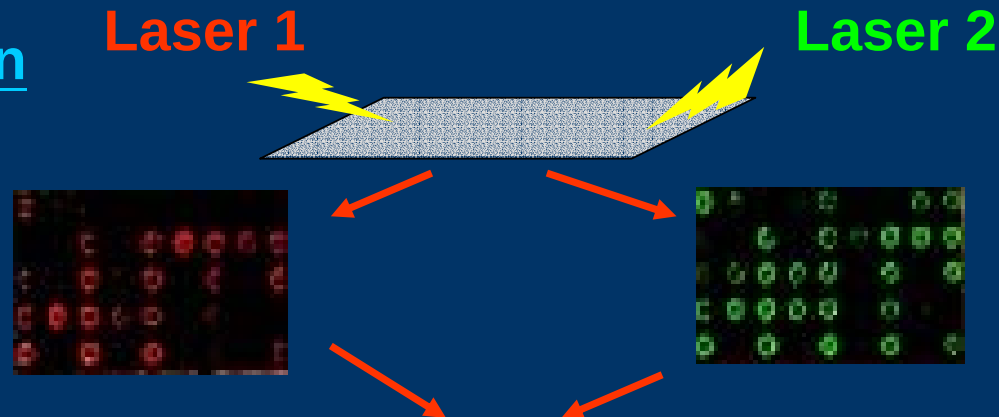


1. Fabrication of arrays

2. Sample preparation

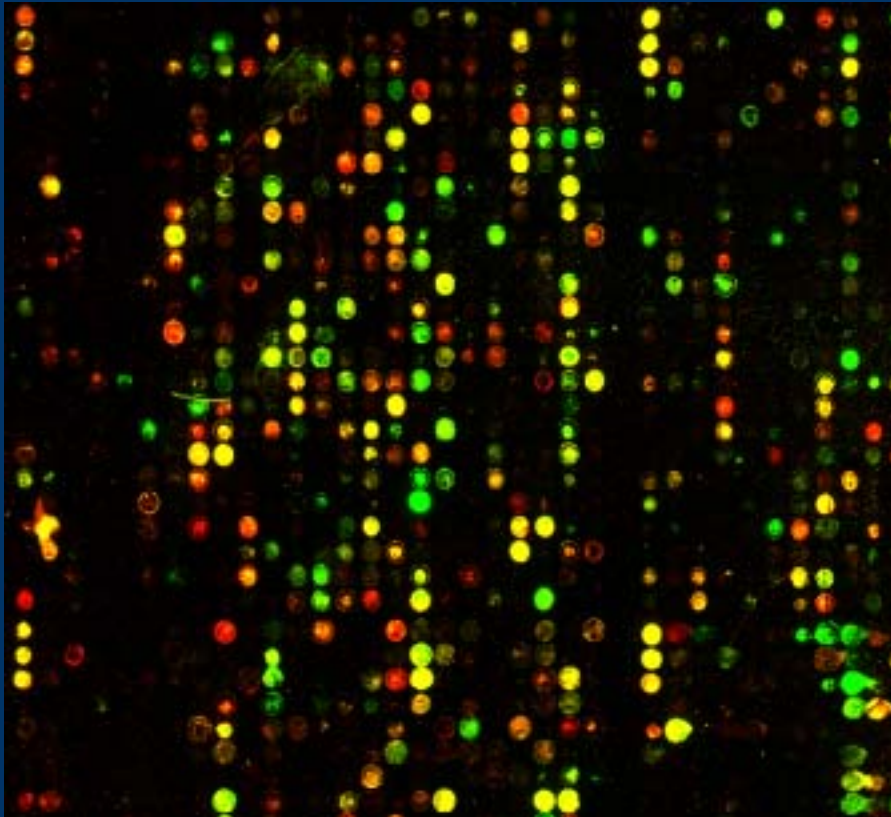
3. Hybridization

4. Detection



5. Data analysis

Pseudo-Color Plot of Differential Gene Expression (Overlay)

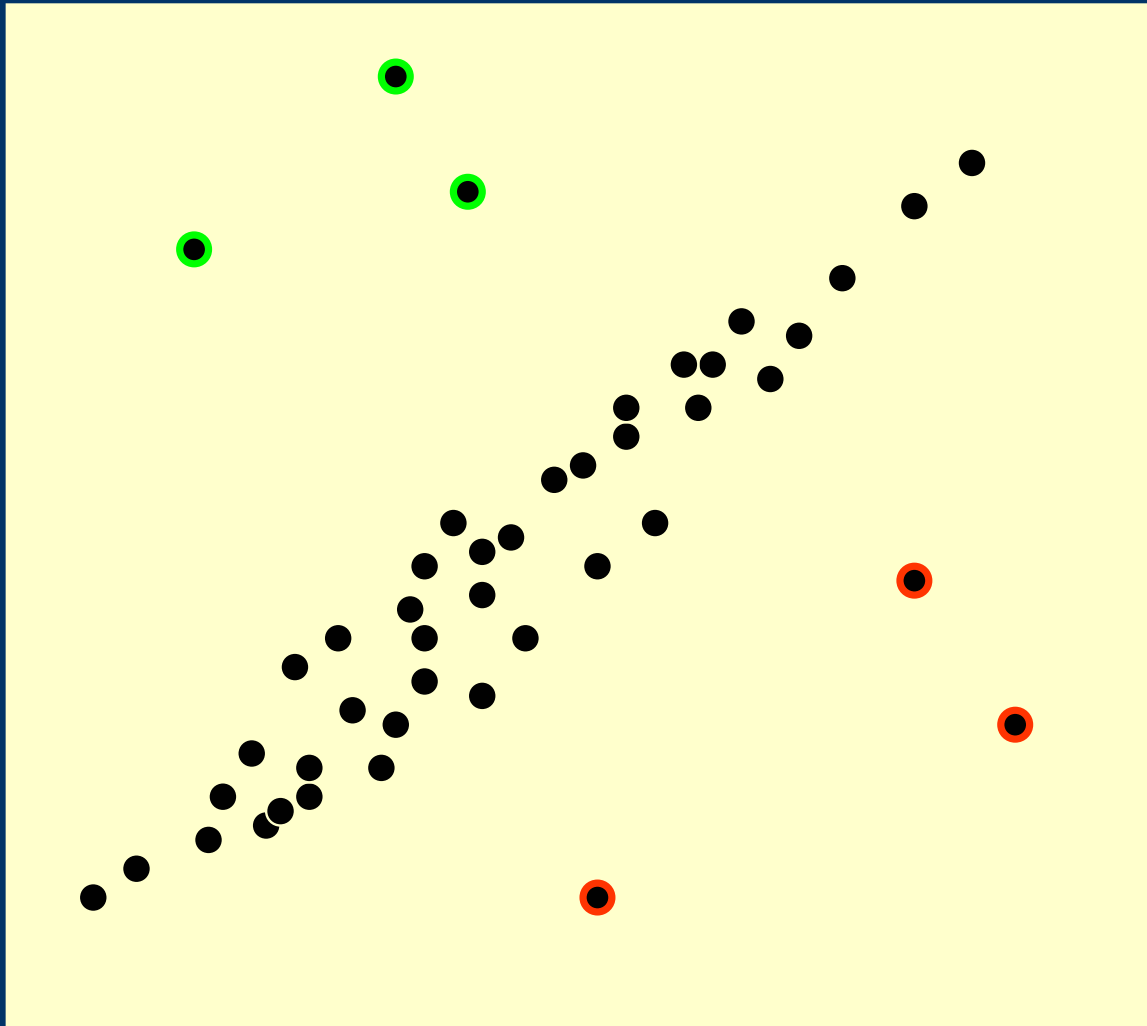


Red: genes differentially expressed gene in sample 1

Green: genes differentially expressed in sample 2

Yellow: genes that are not differentially expressed between sample 1 and 2

Gene Expression Level (Sample 2)



Gene Expression Level (Sample 1)

Data analysis

QuantArray or
ImageOne™

Raw signal intensities, quality factors,
Cy5/cy3 ratios for each genes

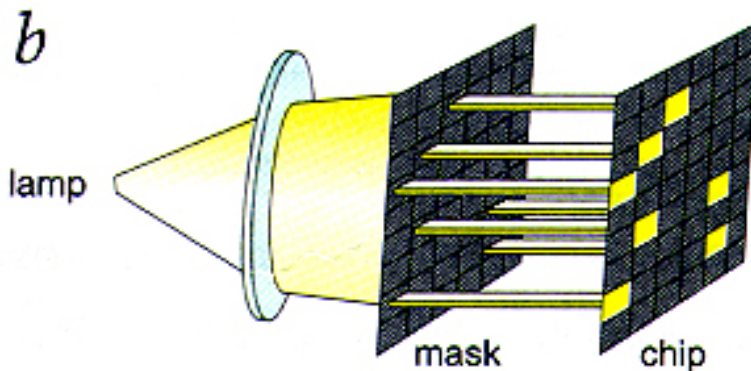
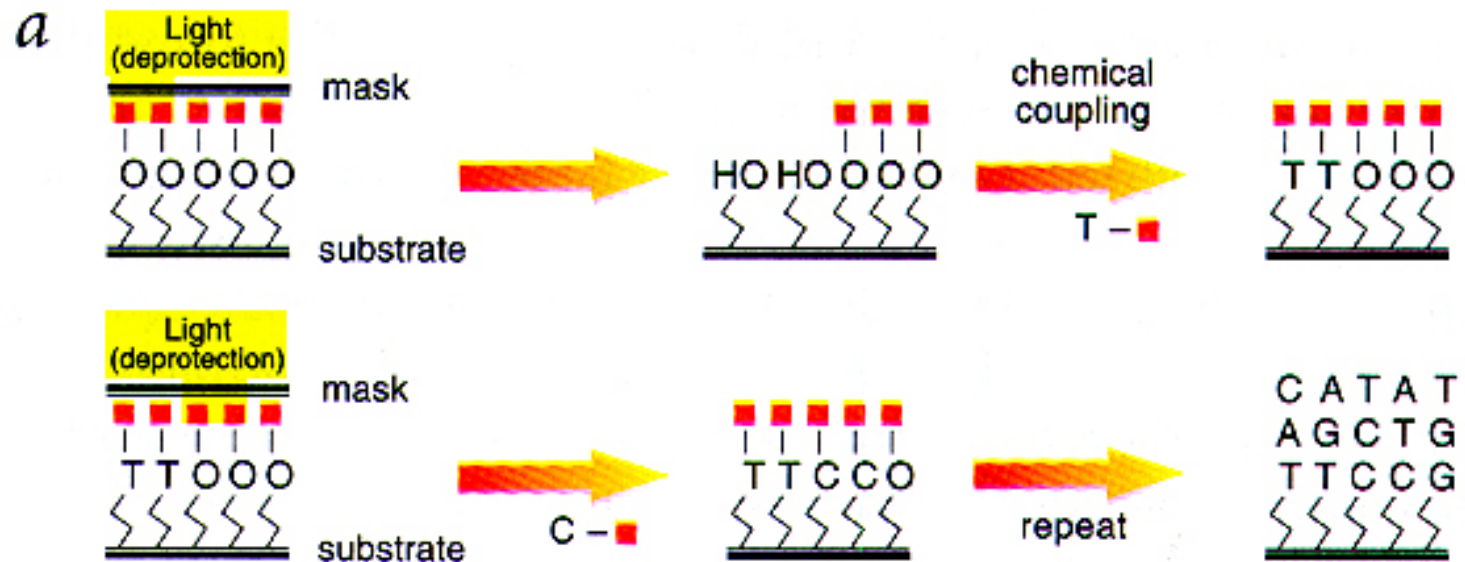


Genesight, GeneSpring, Spotfire, OmniViz™

Data visualization, Clustering
analysis, Logistic queries

Final data

Affymetrix Technology: Light Directed Oligonucleotide Synthesis



From DNA Sequences to Functions

- Expression profiles (transcriptome)
- Protein profiles (proteomics)
- Protein-protein interactions
- Computational analyses and modeling

Proteomics

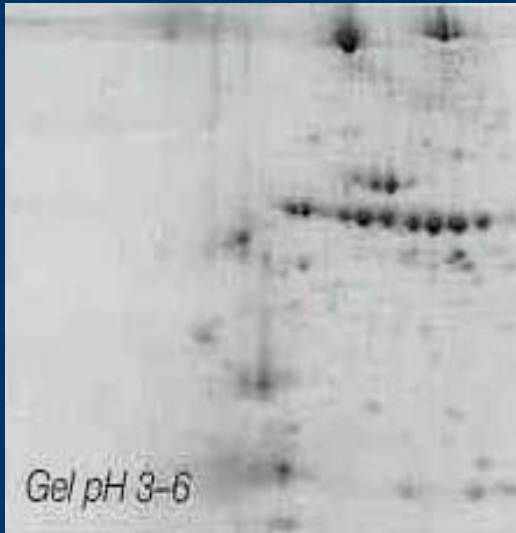
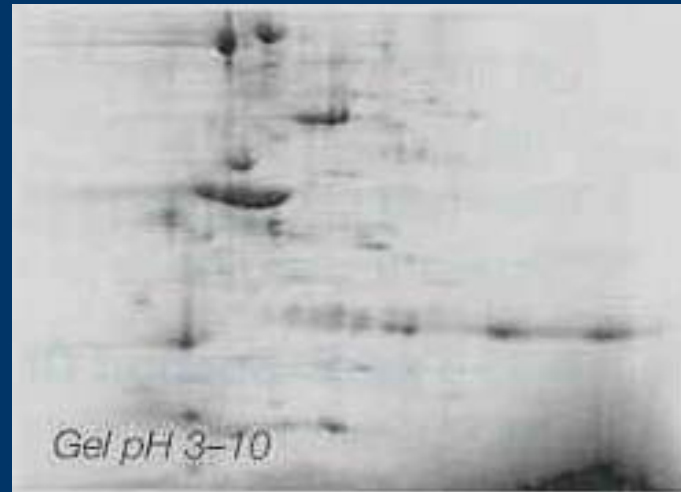
- Classical definition
 - 2-D gels of cell lysates and annotation
 - 2-D gels to visualize differential protein expression

2-D gel Electrophoresis

Zoom gels

3 ← pH → 10

MW



J. Malone, Monsanto Inc.

2-D Gel Protein Profiling

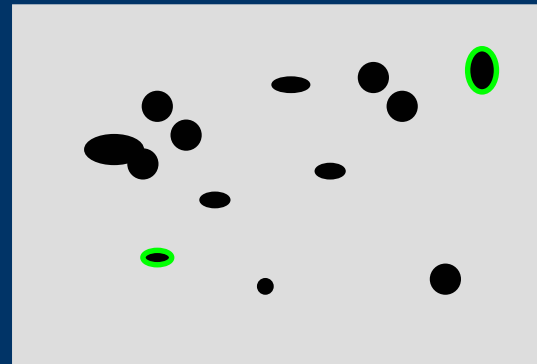
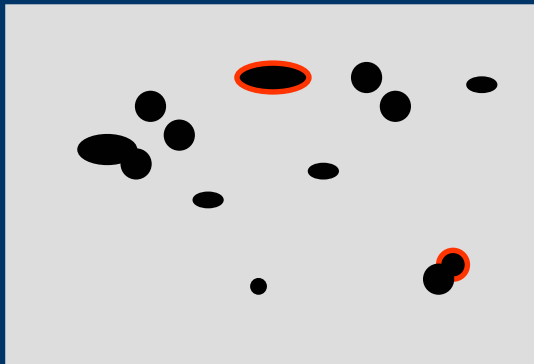
COLLECT AND HARVEST SAMPLES

Designing experiments to profile conditions/perturbations/mutations and carefully controlled growth conditions



TOTAL PROTEIN ISOLATION

Protein yield and purity are determined by system. Protein samples were separated on different 2-D gel. New technology: differentially labelled proteins (e.g. linked to Cy3 and Cy5) run on the same 2-D gel.



Proteomics

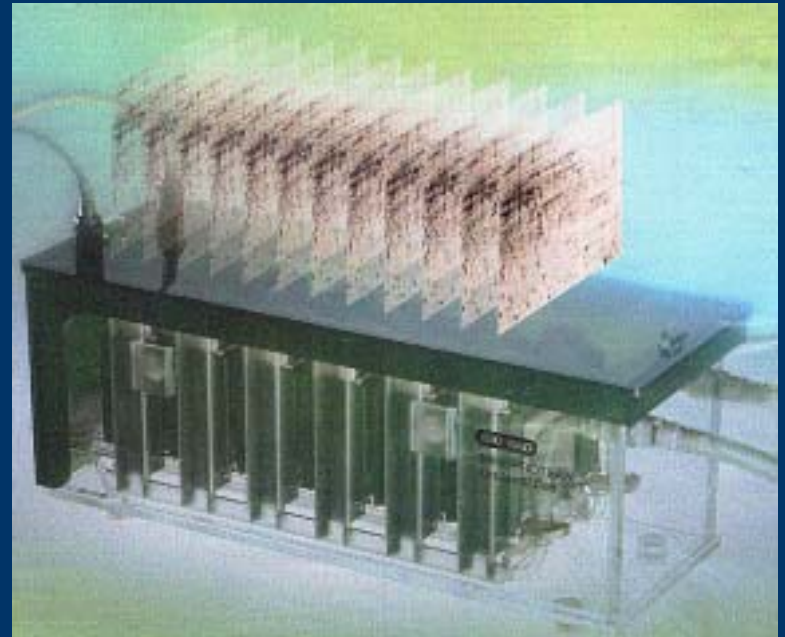
- Post-genomics era definition
 - Protein identification
 - Post-translational modifications
 - Determining function
 - Molecular medicine
 - Differential display by 2-D gels
 - Protein-protein interaction

Recent Advancements in Proteomics Technologies

- High throughput 2-D gel electrophoresis
- Mass spectrometry
- Protein array technologies

Advancement in 2-D Gel Electrophoresis

- Zoom gel
- High-throughput
- Image analysis
- Automatic spot cutter
- Downstream link to mass spectrometry for protein identification (e.g. use florescent stains or SilverQuest from Invitrogen)



Limitation of 2-D Gel Electrophoresis

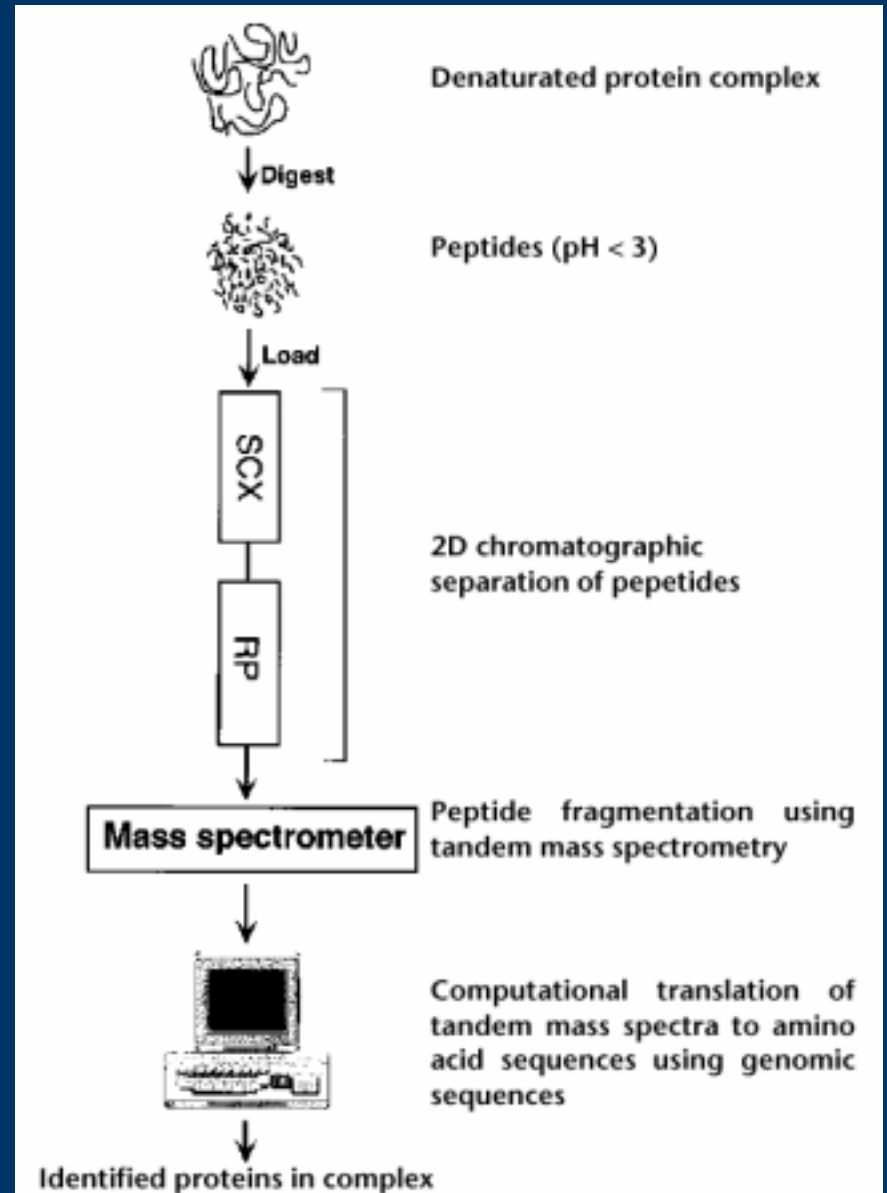
- Not for hydrophobic, especially integral membrane proteins
- Limited by the pH range
- Not easy for low abundant proteins

Progress in Mass Spectrometry

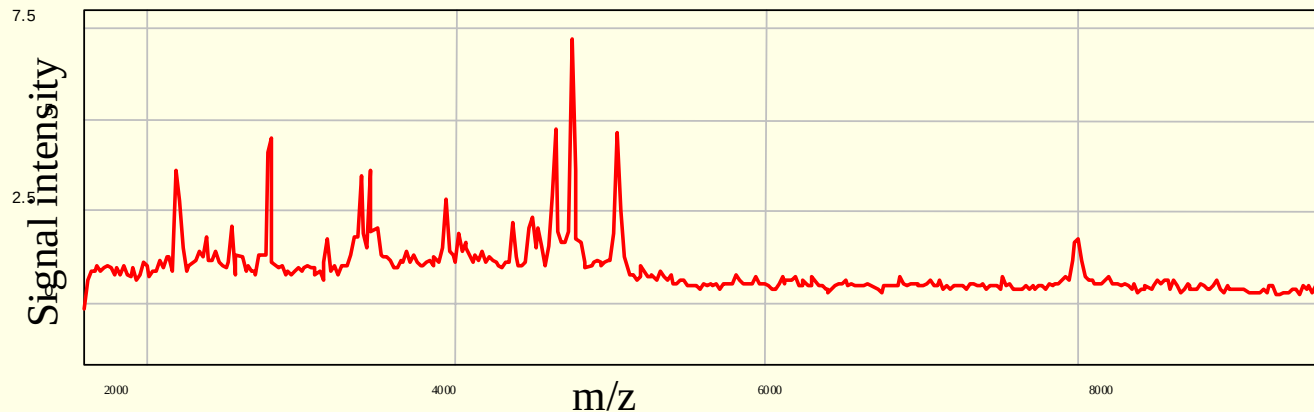
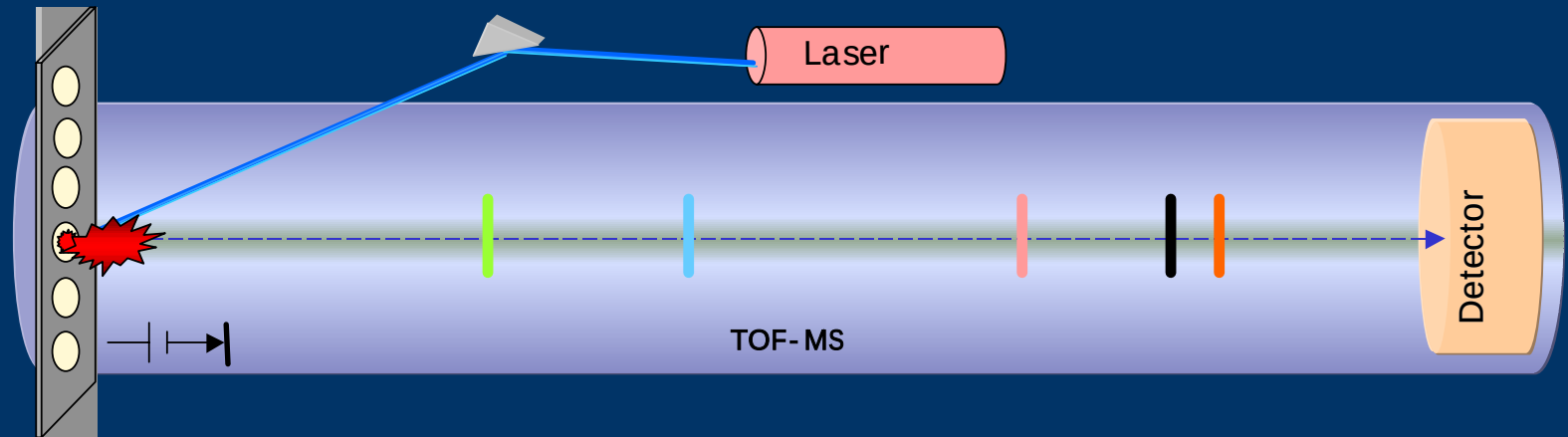
- Multi-dimension liquid chromatography
- Matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectrometry detects high MW proteins
- Protein identification by using tryptic mapping
- Protein sequencing by using mass spectrometry
- Quantitative analysis of proteomes or phosphoproteins

Multi-Dimension Liquid Chromatography (LCMS)

SCX: strong ion exchange
RP: reverse phase

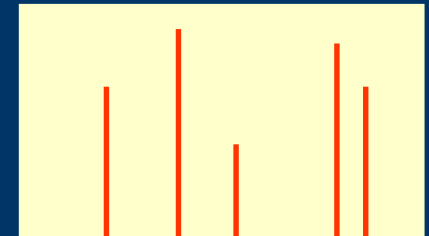


MALDI-TOF Mass Spectrometry



Protein Identification by Tryptic Mapping

Protein $\longrightarrow$ Proteolytic Peptides $\longrightarrow$ Mass Spectrum



M/Z

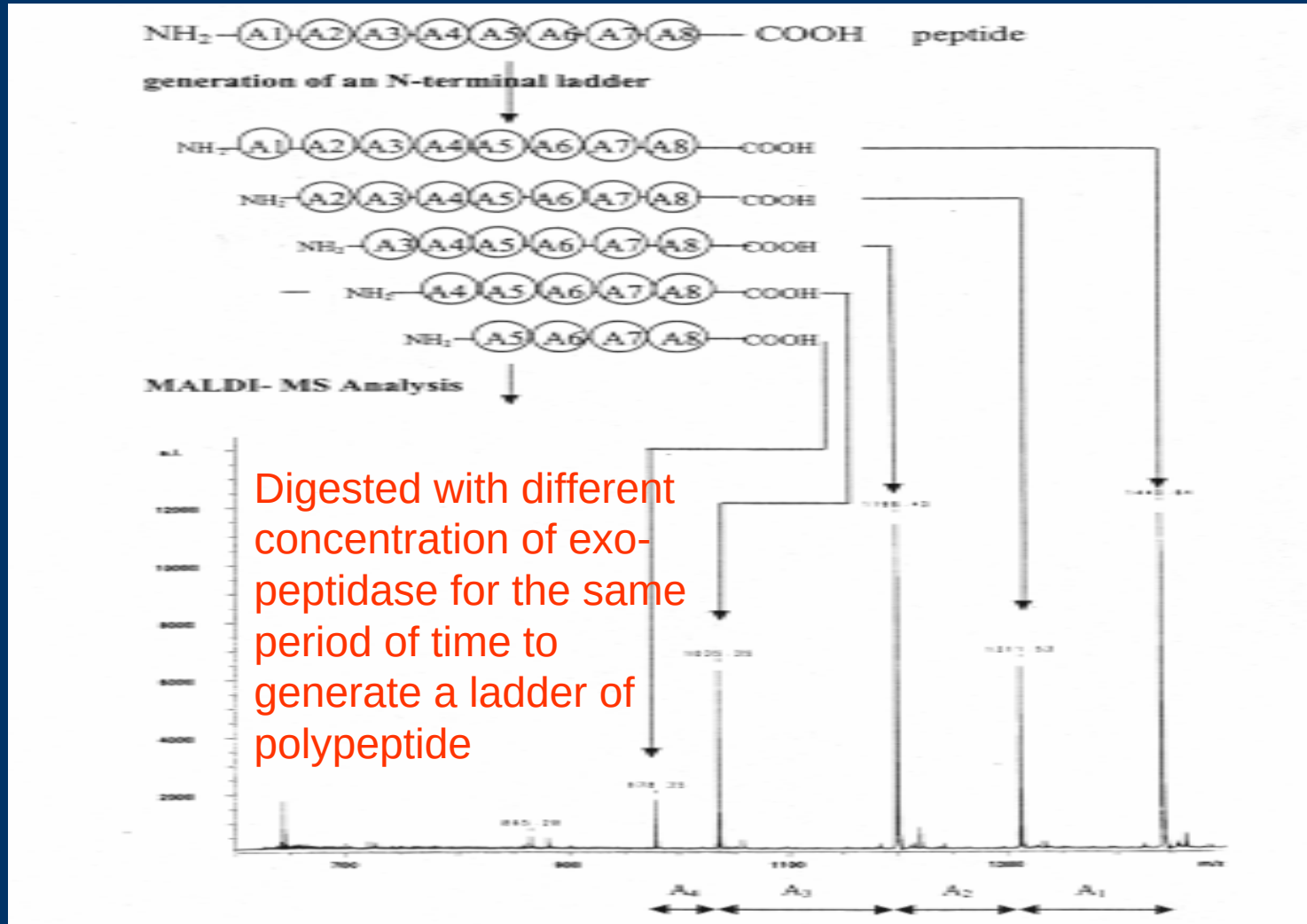
Compare to Database

Known Proteins or
Proteins Predicted
by DNA Sequences

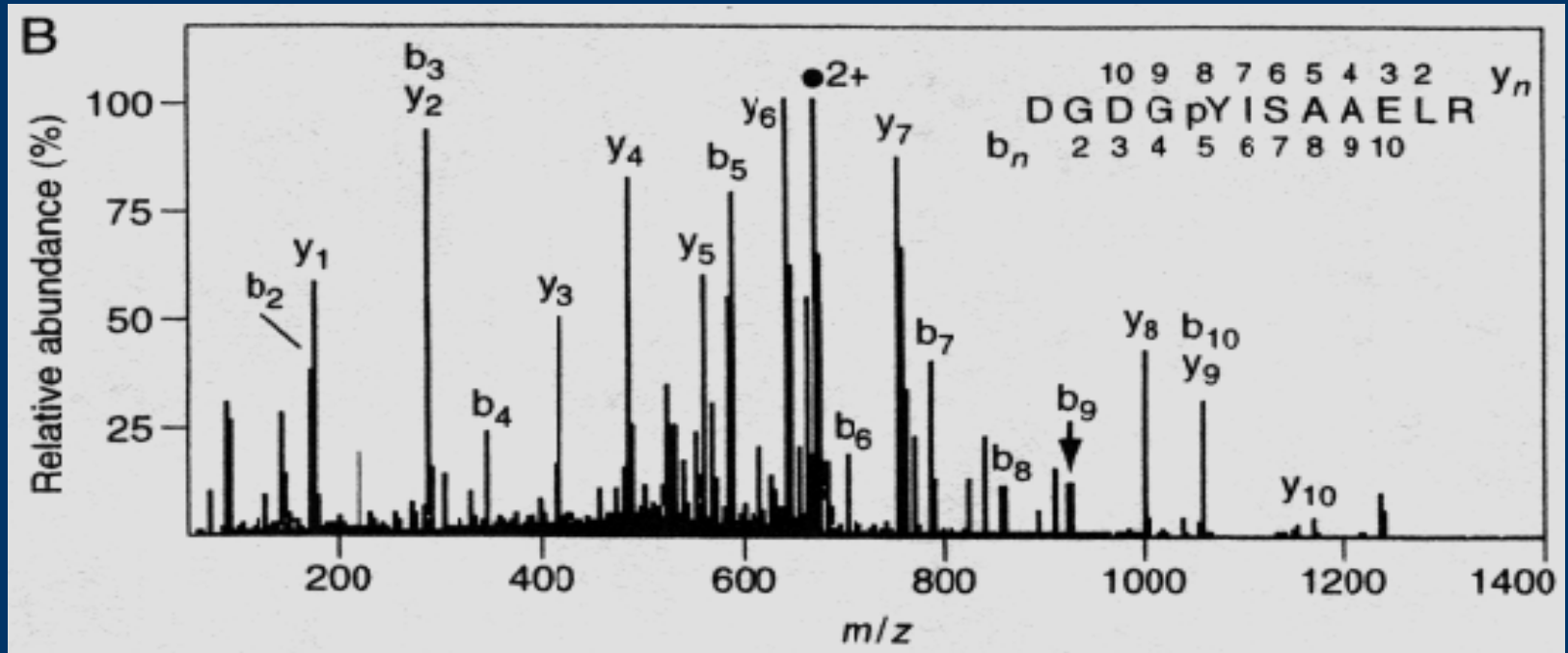
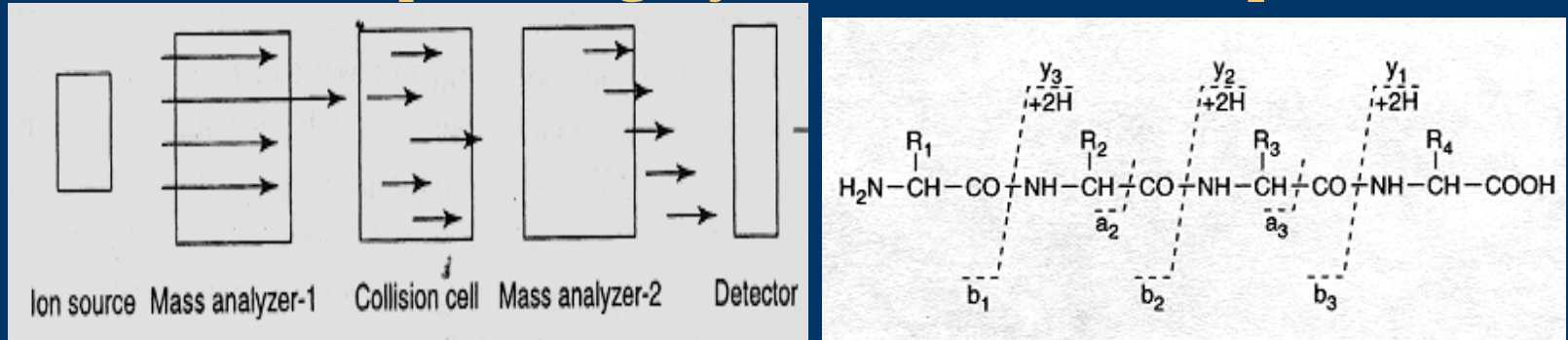
Theoretical
Proteolytic
Peptides

Theoretical
Mass
Spectrum
(Database)

Ladder Sequencing by Mass Spectrometry

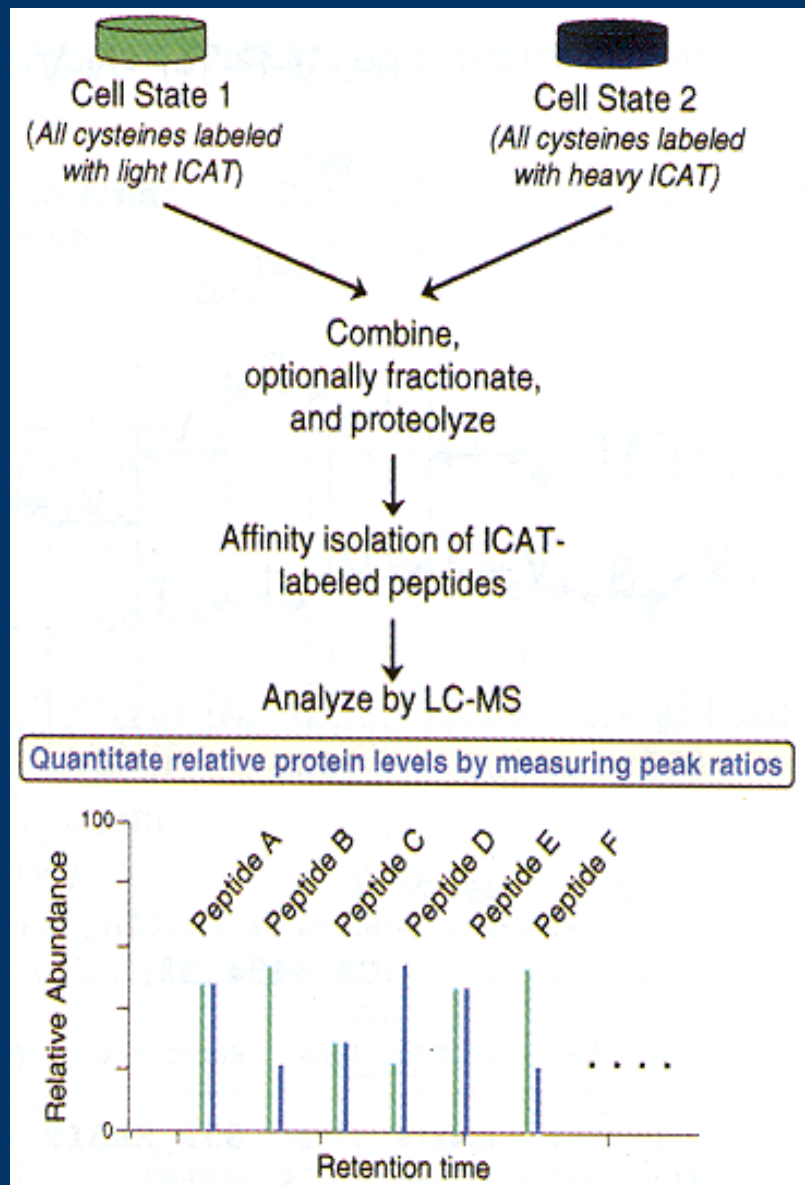
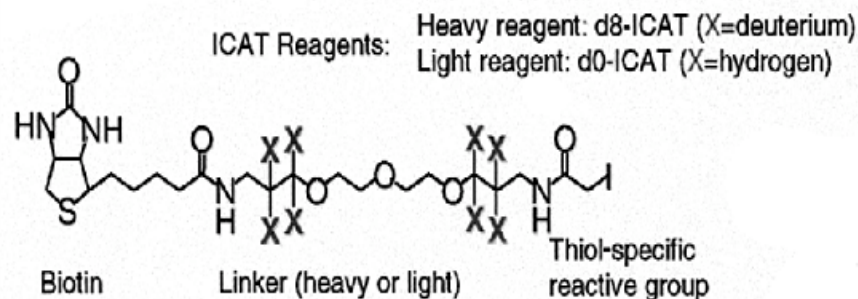


Protein Sequencing by Tandem Mass Spectrometry



(Remark: In Chemical Assisted Fragmentation: only y ions are detected)

Quantitative Proteomics Study Using Isotope- Coded Affinity Tags (ICAT)



How to Build a Function Protein Network

- Direct detection of protein-protein interaction by **yeast two-hybrid**, **protein arrays**, etc.
- Use “guilt-by-association” principle
 - **Phylogenetic profiles**: inheritance of pairs of proteins into various species
 - **Rosetta Stone method**: domain in fused protein versus separate proteins
 - **Gene neighbor method**: if in several genomes the genes that encode two proteins are neighbors on the chromosomes, the proteins tend to be functionally linked

Phylogenetic Profiles

Proteins	Organism		
	1	2	3
A	+	+	-
B	+	-	+
C	-	+	+
D	+	+	+
E	+	+	-
F	-	+	+
G	+	+	-

	1	2	3
A	+	+	-
E	+	+	-
G	+	+	-
C	-	+	+
F	-	+	+

Proteins A, E, G are functionally linked;
Proteins C, F are functionally linked

Rosetta Stone Method

Rosetta Stone in
Organism 1



Protein A in
Organism 2



Protein B in
Organism 2



Proteins A, B are functionally linked

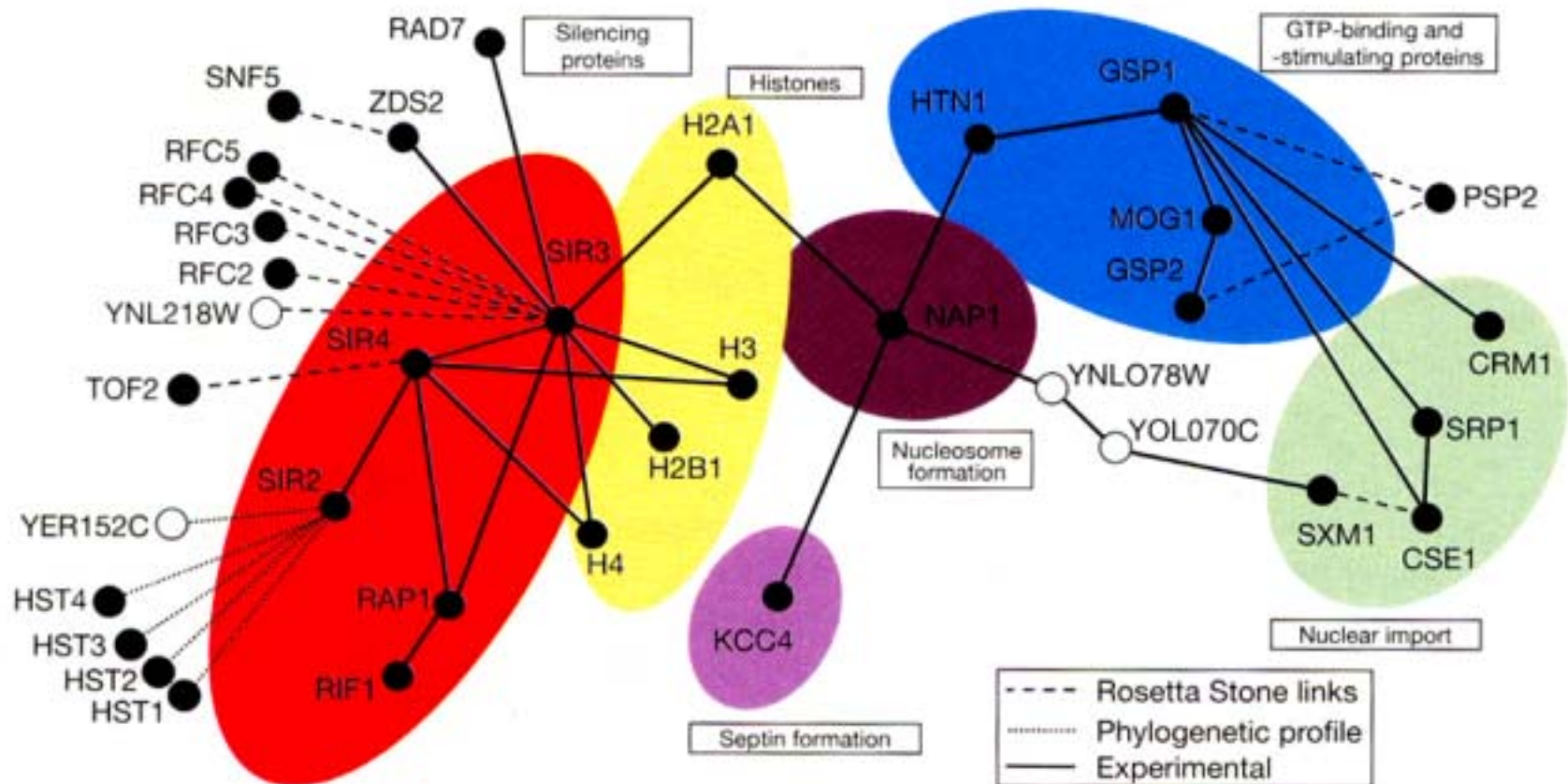
Gene Neighbor Method



Gene 1 and Gene 2 (Protein 1 and Protein 2)
are functionally linked;

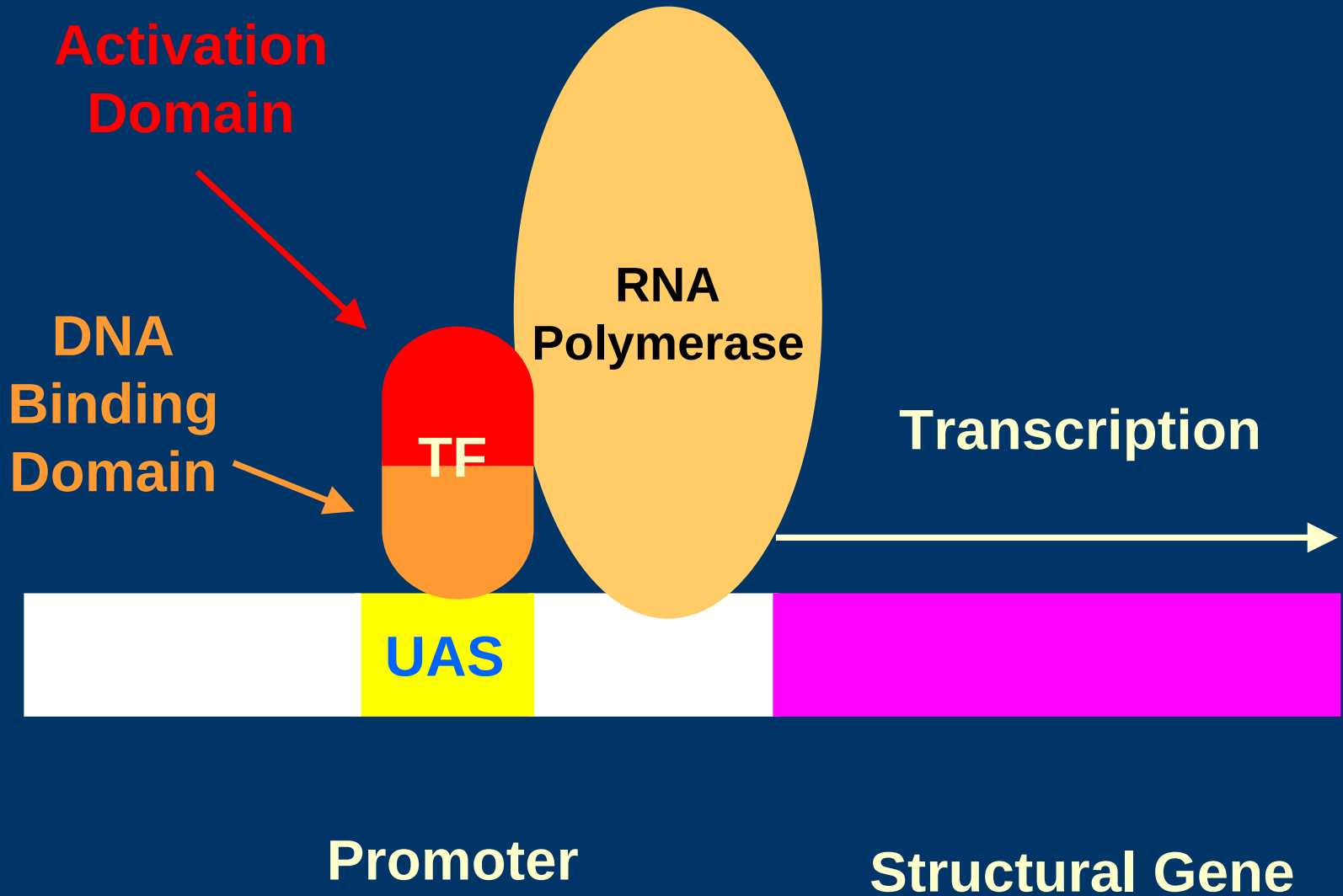
Network of Protein Interactions in Yeast

Eisenberg *et al.*, 2000



From DNA Sequences to Functions

- Expression profiles (transcriptome)
- Protein profiles (proteomics)
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**Activation
Domain**

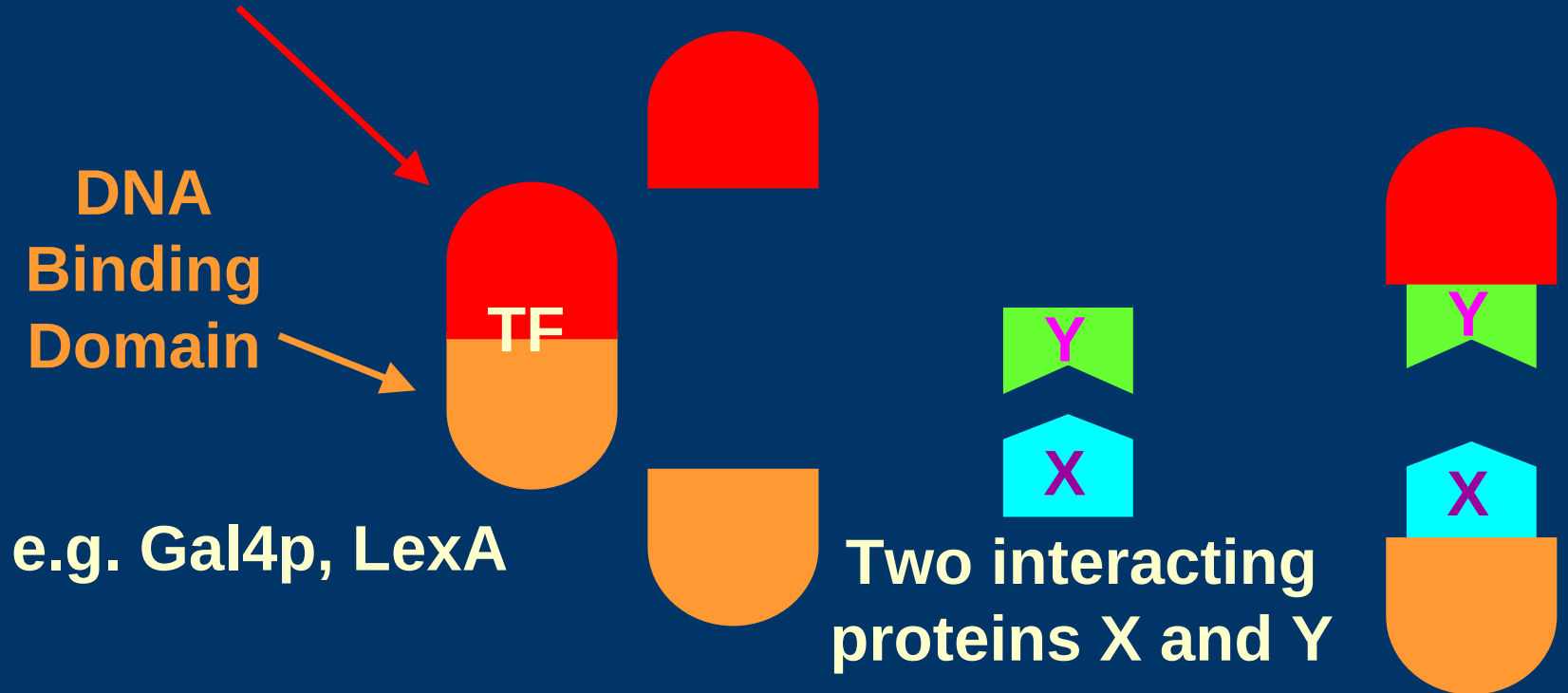
**DNA
Binding
Domain**

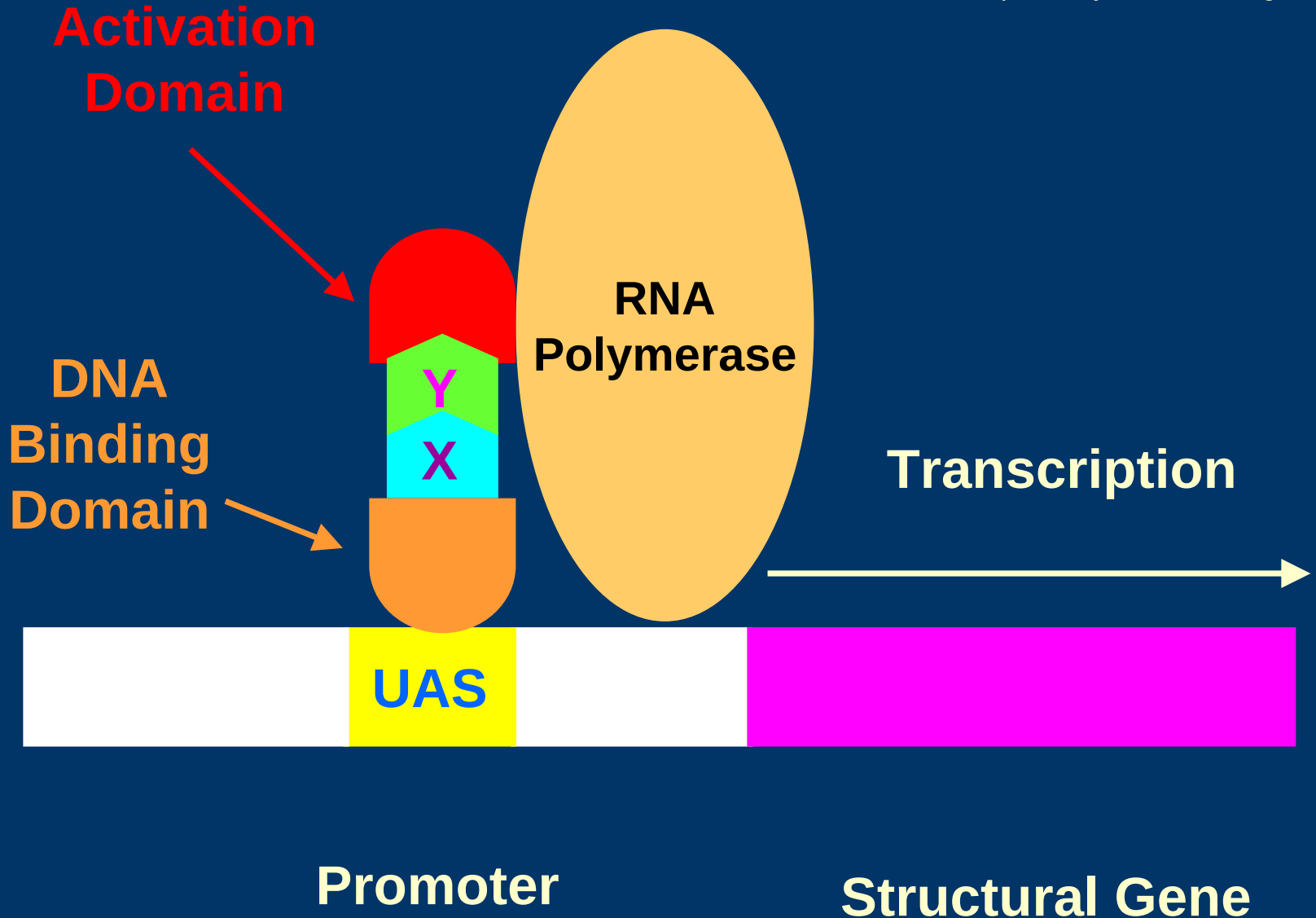
e.g. Gal4p, LexA

Each domain
cannot function
alone

Two interacting
proteins X and Y

**Fusion
Proteins**





Screening for Clones of Interacting Proteins

- Construction of bait
- Transformation into a reporter strain
- Transformation of cDNA library into the reporter strain
- Selection and screening using double markers
- Selection for loss of plasmids containing the bait
- Mating assay to eliminate false positive
- Plasmid recovery and retesting
- Secondary assays

**Positive
Selection
Marker
For
Vector**

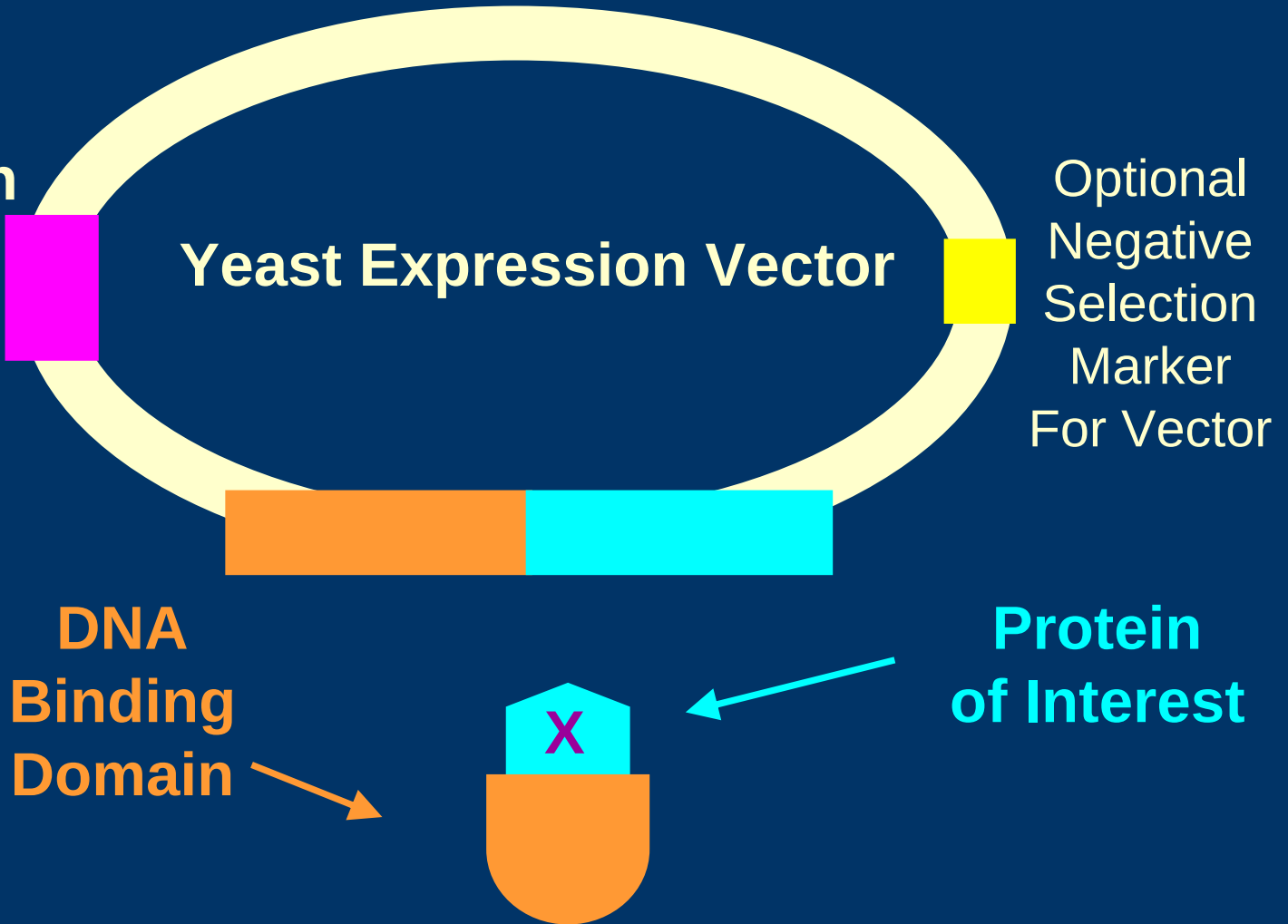
Yeast Expression Vector

**Optional
Negative
Selection
Marker
For Vector**

**DNA
Binding
Domain**

**Protein
of Interest**

**Fusion
Protein (Bait)**



Screening for Clones of Interacting Proteins

- Construction of bait
- Transformation into a reporter strain
- Transformation of cDNA library into the reporter strain
- Selection and screening using double markers
- Selection for loss of plasmids containing the bait
- Mating assay to eliminate false positive
- Plasmid recovery and retesting
- Secondary assays



Promoter

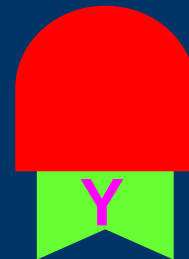
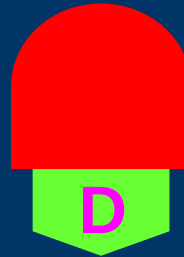
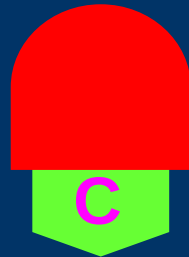
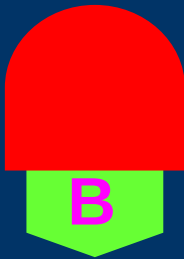
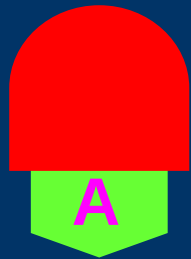
Double Markers
e.g. *URA3*, *lacZ*

Screening for Clones of Interacting Proteins

- Construction of bait
- Transformation into a reporter strain
- Transformation of cDNA library into the reporter strain
- Selection and screening using double markers
- Selection for loss of plasmids containing the bait
- Mating assay to eliminate false positive
- Plasmid recovery and retesting
- Secondary assays

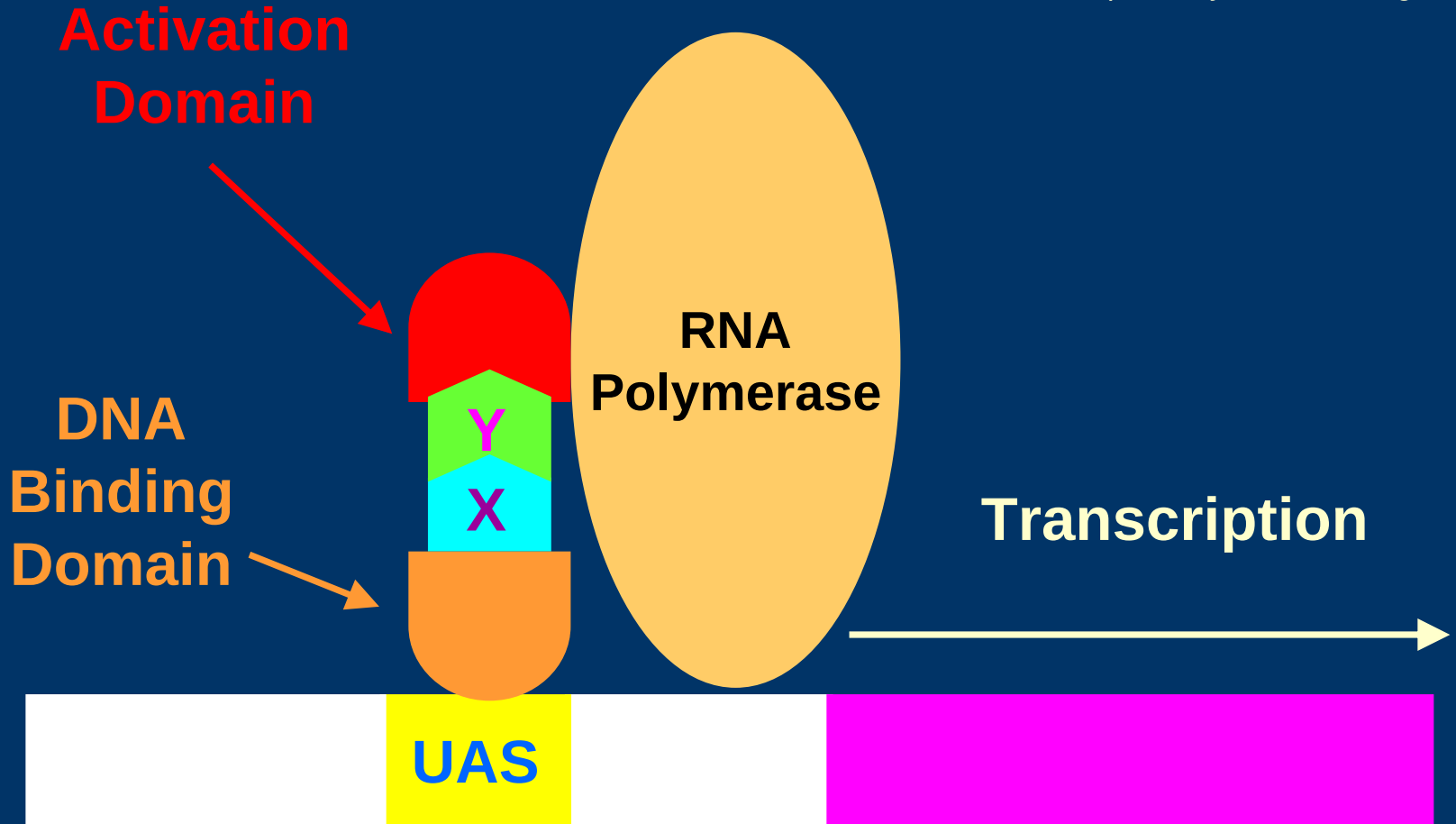
**Positive
Selection
Marker
For
Vector**

Yeast Expression Vector



**Interacting
Protein
(prey)**

A Library of Fusion Proteins



Double Markers

e.g. *URA3* for survival on uracil minus medium
lacZ for blue-white screening with X-gal

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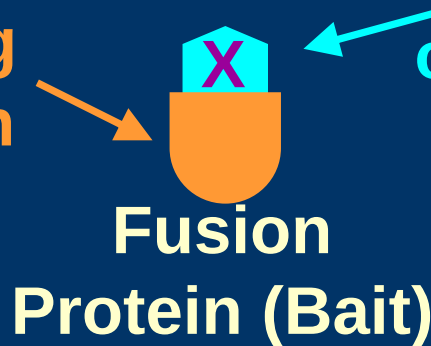


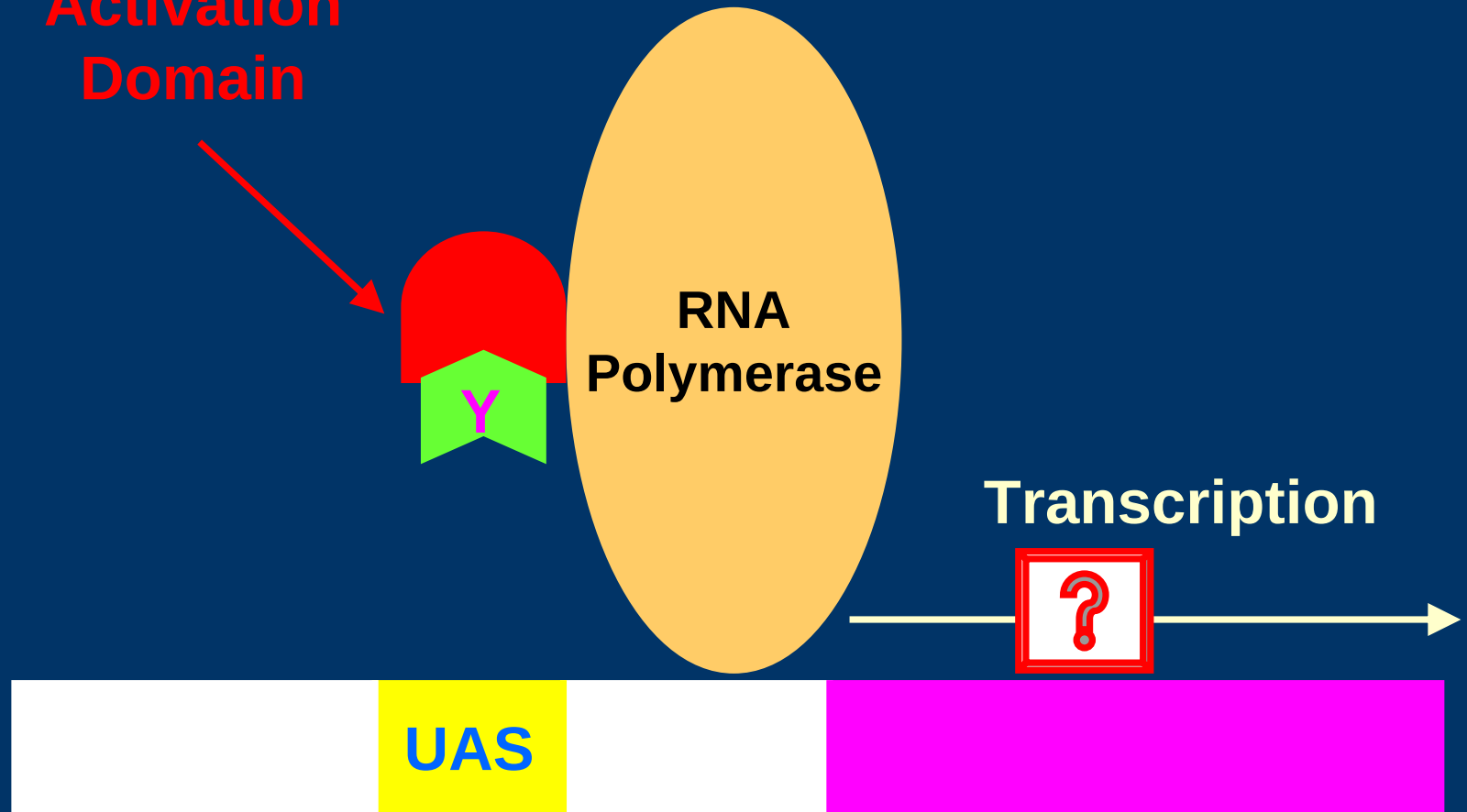
CYH2
(cycloheximide
sensitive)

The yeast
host contains
a mutation in
CYH2 gene and
hence is resistant
to cycloheximide

DNA
Binding
Domain

Protein
of Interest





Double Markers

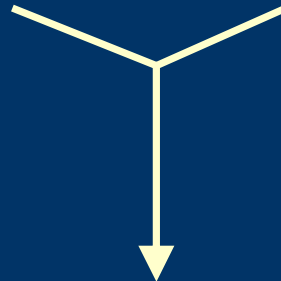
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Screening for Clones of Interacting Proteins

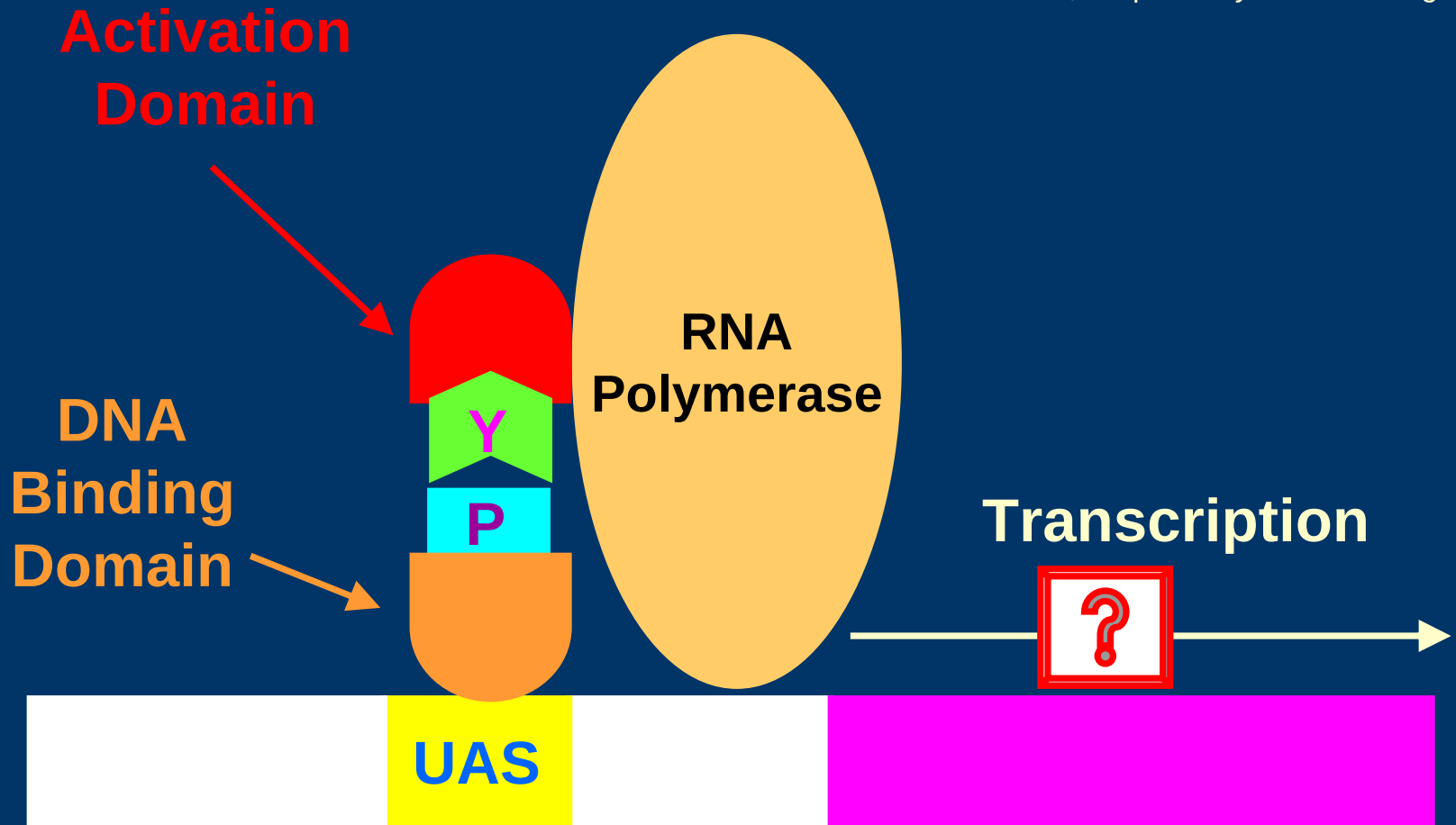
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**Yeast making
Y-Gal4 activation domain
fusion protein**

**Yeast making
Gal4-P fusion protein
(P = any bait other than X)**



**Check for false positive
using the resulting
diploid yeast**



Double Markers

e.g. *URA3* for survival on uracil minus medium
lacZ for blue-white screening with X-gal

Screening for Clones of Interacting Proteins

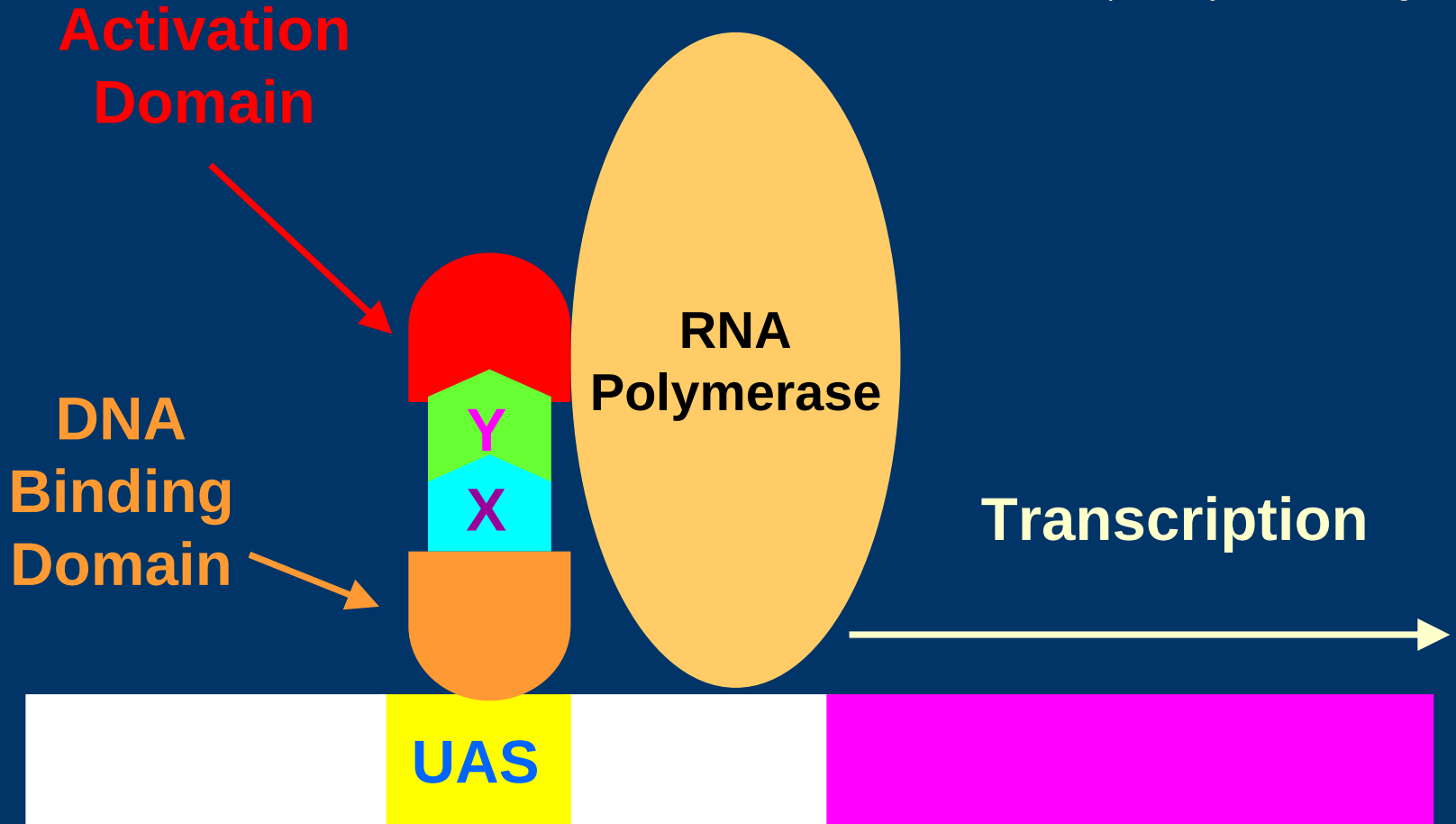
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Secondary Assays

- Make radioactively-labeled Y (prey) by *in vitro* transcription-translation coupling, run through affinity column of X (bait)
- Co-immunoprecipitation
- Switching the bait and prey in their respective plasmids

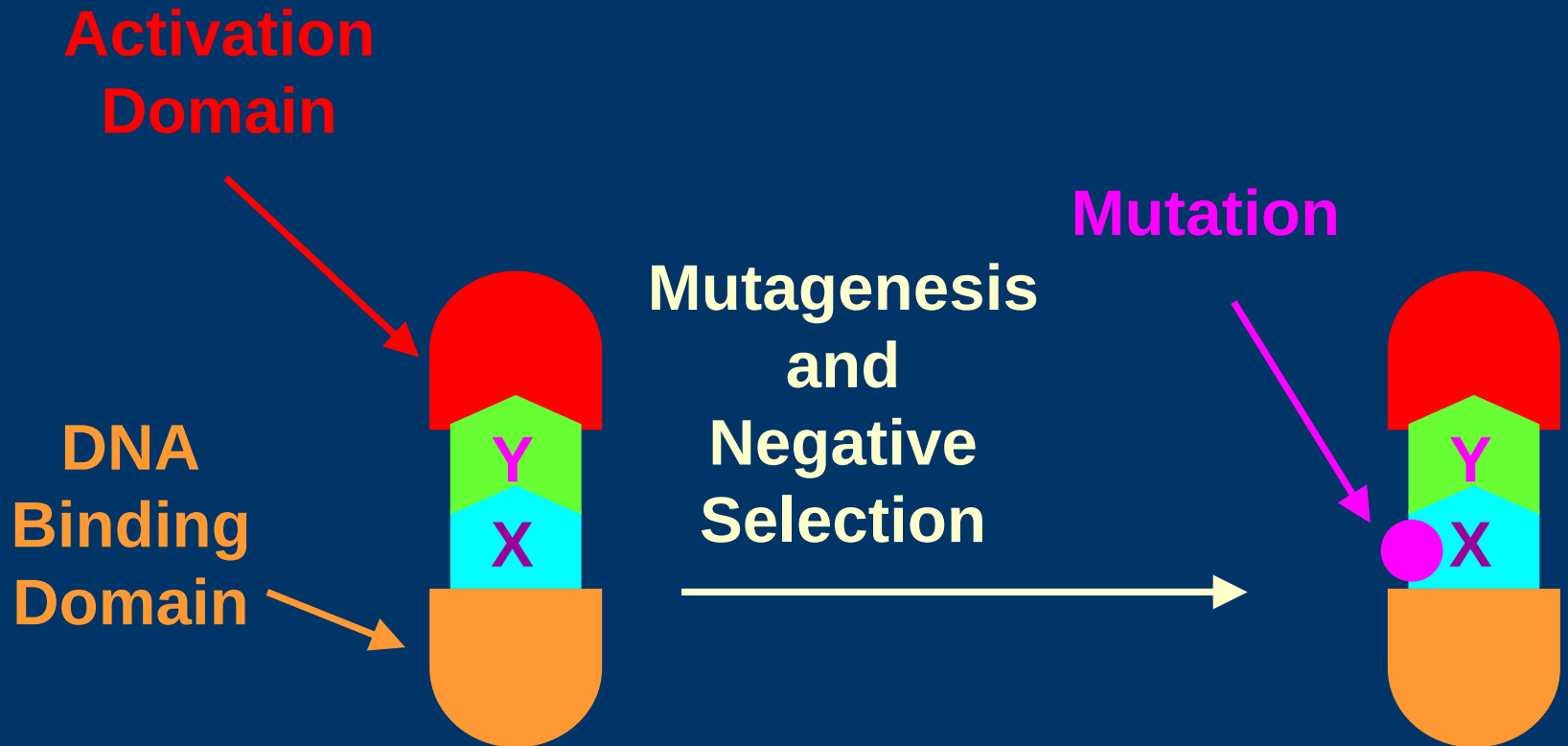
Reverse Two-Hybrid System

- Allow genetic selection of events responsible for the dissociation of particular protein-protein interaction;
- Important for dissecting interacting domains

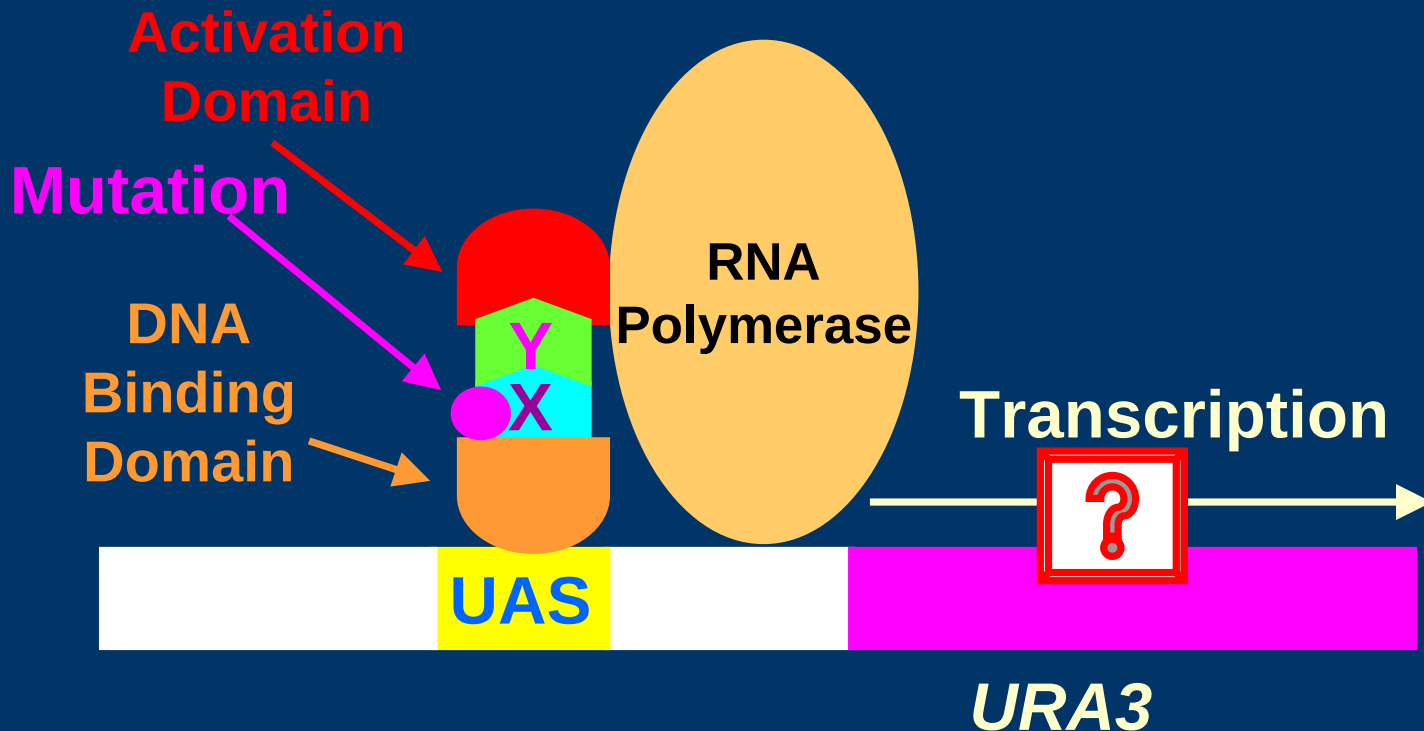


Double Markers

e.g. *URA3* for survival on uracil minus medium
lacZ for blue-white screening with X-gal



An Example of Negative Selection



URA3 is a dual reporter gene: positive selection on uracil minus media and negative selection on Foa (fluoroorotic acid) containing media